PHINMA UNIVERSITY OF ILOILO

College of Information Technology Education

CAPSTONE PROJECT PROPOSAL

Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics

In partial fulfillment for the requirements in Capstone 1 and Research Subject

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CHAPTER 1

1.1 Background of the Study

In today's fast-paced world, businesses across all sectors are shifting to digital platforms to keep up with the growing expectations of modern consumers. The food service and baking industry in the Philippines is experiencing this exact transition, as tech-savvy individuals increasingly rely on mobile devices and computers to browse menus, place orders, and settle transactions remotely. Because of this shift in consumer behavior, small local micro-enterprises that continue to rely on manual, paper-based processes face severe operational challenges. Traditional, non-automated methods often lead to critical inaccuracies, limited market reach, and an inability to meet daily customer expectations. For small community bakeries, operating without a modern web presence means losing potential sales to larger competitors who offer instant online ordering, clear pricing, and transparent communication.

The core issue stems from how order information is captured, stored, and processed during daily operations. In a digital commerce platform, data entries are structured, validated, and processed automatically by backend logic. Conversely, traditional store setups rely heavily on loose paper slips, handwritten notebooks, and human memory. This lack of automated validation leaves daily transactions highly vulnerable to misplaced records, delayed order tracking, and operational confusion. When a small business operates entirely without digital validation, everyday tasks like taking an order, calculating a price, or scheduling a delivery turn into stressful, repetitive manual chores that consume huge amounts of time and lower staff productivity. For community-scale enterprises, continuing to operate within a purely analog framework creates a technical barrier that cuts them off from the modern digital marketplace, making it increasingly difficult to sustain consistent growth or build long-term customer loyalty.

C&N Bakery is a local establishment serving a growing customer base within a competitive community market. The bakery offers a wide selection of daily baked goods, including fresh breads, pastries, cakes, and traditional local delicacies . In addition to selling standard retail items over the counter, the bakery specializes in custom cake orders made for special events like birthdays, weddings, debuts, and anniversaries. These custom orders represent a significant portion of the business's overall revenue and require careful planning and precise execution. Despite managing a diverse product menu and handling many daily customers, C&N Bakery handles its entire workflow through manual mechanisms, such as walk-in counter visits, phone calls, and scattered social media messages on Facebook. The business operates without a centralized digital system to organize its daily activities, leaving crucial customer choices and transaction records scattered across different notebooks, paper slips, and informal message threads.

When a business operates without a single central system to store its records, information easily gets lost or fragmented. At C&N Bakery, a single custom transaction often leaves a scattered data trail across multiple communication channels. Design layouts, delivery deadlines, special instructions, and payment receipts are frequently recorded in completely different places. Because these details are isolated from one another, staff members must manually piece together the customer's requirements before baking can start. This manual approach consumes massive amounts of time and makes day-to-day operations chaotic, turning simple routine tasks into exhausting manual procedures that drain the small business's daily energy and workforce productivity.

Managing custom cake orders is inherently far more complex than selling pre-baked goods over the counter. A single custom cake request requires capturing and tracking multiple precise attributes simultaneously, including cake dimensions, tier structures, flavor variations, frosting types, color schemes, personalized dedication messages, and visual reference photos. At the same time, the bakery must carefully coordinate pickup or delivery dates to guarantee product freshness and structural integrity during transit. Because C&N Bakery currently relies on handwritten logs, loose paper receipts, and unorganized chat histories, staff members can easily misread or forget specific design preferences by the time production begins in the kitchen. Furthermore, writing down fulfillment dates manually in physical logbooks frequently causes human error, resulting in accidental double-bookings that overload the kitchen team beyond their daily shift capacity.

This operational vulnerability is deeply tied to the problem of manual transcription. When a customer outlines their custom selections through casual communication channels, that information exists as an unformatted text string. As that request moves from the front counter staff to the production area, the information is manually rewritten onto paper order tags or kitchen whiteboards. Every time a human manually copies, interprets, or rewrites an order detail, the risk of human error occurring increases significantly. Essential design choices can easily be misread, miscalculated, or completely forgotten during the transfer. This manual workflow creates a weak operational loop where the original intent of the customer rarely matches the actual physical instructions given to the baking staff on the kitchen floor.

Every time an order detail is manually transcribed, the risk of miscalculation increases. Essential design preferences can easily be misinterpreted, miscalculated, or left out entirely. When a cake is prepared with incorrect specifications, it triggers a chain reaction of operational failures across the business, resulting in wasted raw ingredients, disrupted baking schedules, lost revenue, and damaged customer trust. The financial impact of these errors is particularly severe due to the rigid nature of the baking process itself. Unlike quick-service meals that can be replaced in minutes, custom cakes require hours of uninterrupted baking, cooling, and detailed decorating. If an error is noticed late in the process, the entire cake often has to be discarded, delaying every other order waiting in the kitchen queue and creating a backlog that ruins the entire work shift.

These internal difficulties are further magnified by the rapid adoption of digital payment solutions in the Philippines, particularly electronic wallets like GCash and online bank transfers. Modern consumers expect clear pricing, immediate feedback, and simple remote payment options. At C&N Bakery, operating manually creates major communication bottlenecks. When a customer sends a message asking for a price quote on a custom cake design, kitchen staff must pause their physical work to manually calculate costs line-by-line based on cake size, tiers, and add-ons. This manual step leads to long response delays, leaving potential buyers waiting in digital queues for simple information. In an era where buyers expect instant answers, waiting hours for a message reply causes many customers to abandon their inquiries and purchase from larger, automated competitors instead.

Furthermore, once an order is placed, customers currently lack a clear way to monitor their order status in real time. Because there is no online order tracker, customers are forced to call or message the store repeatedly to ask if their cake is being prepared, baked, or ready for pickup. This constant back-and-forth communication gap creates a major operational barrier inside the store. Front counter staff and kitchen personnel do not share a synchronized information system, making it hard to track order priorities and preparation sequences. Staff members must constantly stop their physical duties to verbally verify order statuses with the bakers. This constant workplace interruption wastes valuable labor time, stresses out the staff, and leads to scheduling mistakes that leave waiting customers frustrated.

Operating without a secure web system also introduces serious data security risks and administrative obstacles. On the customer side, users lack a secure account interface to review their past orders or monitor active deliveries. For administrative personnel, operating without a secure, centralized administrative dashboard means keeping financial records, customer contact details, and transaction histories in paper notebooks stored directly inside the active kitchen. These paper logs are constantly exposed to water spills, flour dust, grease stains, heat degradation, and physical loss. If a single notebook is misplaced, damaged, or destroyed, the bakery permanently loses its operational records with no way to recover the data. Storing customer addresses and phone numbers on loose paper slips also presents privacy concerns, while management is left without historical records to evaluate monthly revenue, track sales growth, or identify best-selling product trends.

To resolve these operational vulnerabilities, this study will develop the Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery. The proposed web application will establish a centralized digital platform engineered to streamline order taking, dynamically balance daily kitchen capacity limits, track production stages across a clear lifecycle, and compile historical records into clear visual sales reports. The system will provide dedicated user accounts for registered customers while maintaining open browsing access for guest users.

Through this proposed platform, customers will be able to customize their cake selections, view automated cost calculations, submit reference references, and select available fulfillment dates backed by real-time schedule checks to prevent kitchen overbooking. Administrative staff will be provided with a centralized management dashboard to verify order payments, update production progress in real time, and view visual sales analytics. Built using cost-effective cloud database technology, this practical architecture will keep operational expenses manageable, safeguard store transaction records, eliminate manual data entry errors, optimize kitchen productivity, and provide C&N Bakery with an efficient, reliable web presence tailored to its daily business needs.

1.2 Statement of the Problem

1.2.1 General Problem Statement

The primary challenge facing C&N Bakery is its total reliance on manual operational workflows, physical paper logbooks, and fragmented social media chat threads to manage daily business activities. Because the establishment lacks a centralized web platform to sync data across daily tasks, operational records remain severely fragmented. Key order details—including custom cake specifications, pickup schedules, menu updates, raw material availability, and customer records—are scattered across loose paper slips, handwritten kitchen calendars, and unorganized messaging histories.

Relying on this manual setup makes the business highly vulnerable to data entry errors, miscommunication, and permanent loss of physical files. Without an automated digital system to systematically capture, validate, and save order details upon checkout, the bakery operates without a single, verified source of record for its daily transactions. On a daily level, this manual workflow leads to severe communication delays during design inquiries, frequent cake decorating errors on the kitchen floor, costly ingredient waste, and accidental kitchen overbookings. Furthermore, storing financial records, payment logs, and customer contact information in unsecured physical notebooks exposes the bakery to privacy risks and permanent record loss. To resolve these operational vulnerabilities, the proposed Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery will establish a centralized digital platform to automate order processing, sync kitchen schedules, track fulfillment stages, and compile real-time sales records.

1.2.2 Specific Problem Statement

To address the general problem, the specific challenges facing C&N Bakery are categorized into three major operational areas:

1. Customer Intake and Ordering Inefficiencies

- **Communication Backlogs and Delayed Responses:** Customers who want to browse the current menu, check available flavors, verify daily product stocks, or ask about custom cake pricing are forced to call the shop directly or send private messages through social media. Because customer communication relies entirely on manual text replies from staff members who are simultaneously busy mixing batter, decorating cakes, or handling walk-in counter queues, the bakery experiences severe message delays during busy periods, holidays, and weekends. It is common for potential buyers to wait several hours—or even until the next business day just to receive a basic price quote or check if a date is open. This waiting time frustrates interested buyers during critical decision-making moments and causes high customer drop-off rates, as buyers abandon the inquiry process to buy from faster online competitors instead.

- **Pricing Guesswork and Math Errors:** Staff members currently determine prices during messaging conversations by relying on mental math, rough estimates, or checking loose physical price sheets. This informal approach frequently leads to inconsistent price quotes for similar cake designs, human calculation mistakes, and a total lack of itemized cost breakdowns for the buyer. Because there is no structured pricing calculator available to the customer, the buyer cannot see how changing specific design options—such as adding an extra cake tier, choosing a specialty flavor filling, requesting fondant decorations, or setting an urgent delivery date—directly affects their total price. This lack of clear pricing creates consumer hesitation, leads to long price negotiations, and damages customer trust during the ordering process.
- **Misunderstood Cake Designs and Transcription Errors:** Staff members must regularly read through long, unorganized chat

histories across messaging apps to find a customer's custom design requests and manually copy those details onto physical paper kitchen slips. This manual process creates a high risk of misreading or forgetting critical cake specifications—such as exact color themes, tier dimensions, frosting types, specialty toppers, and custom written inscriptions. Because there is no structured form or digital checklist to capture these details during the intake phase, important preferences frequently get buried in casual chat conversations. When these design details are misread or forgotten during the transfer to the production floor, it leads to incorrect cake assemblies in the kitchen. Ultimately, these copying mistakes result in client rejections upon pickup, last-minute re-work, and a heavy waste of expensive baking ingredients that directly eats into profits.

- **Information Gaps and Customer Hesitation:** C&N Bakery's manual setup offers no public self-service section for basic store policies, disclaimers, lead-time rules, or payment guides. Customers seeking basic operational details such as required order advance times, refund rules, allergen warnings, or digital payment instructions—are forced to message the staff directly to ask about them. This lack of clear, accessible information creates unnecessary communication backlogs, forcing workers to answer the same repetitive questions over and over again. More importantly, it causes transaction hesitation during the final stages of ordering, as prospective buyers are often reluctant to send down-payments without seeing clear, official shop guidelines upfront.

2. Operational, Scheduling, and Production Disruptions

- **Kitchen Overbooking and Overlapping Deadlines:** Fulfillment deadlines are currently tracked using a single physical wall calendar hung in the kitchen area. Because multiple employees accept customer orders simultaneously across direct phone calls, walk in counter visits, and messaging channels, workers lack a shared, real time view of the daily production schedule. As a result, staff members routinely book orders without knowing that the kitchen has already reached its maximum workload capacity for a specific date or time slot. This lack of centralized schedule management regularly leads to accidental overbookings and overlapping fulfillment deadlines. On busy days, this overbooking overloads the kitchen team, forcing bakers to rush through complex preparation steps, compromising cake decorating quality, and causing delayed customer pickups that harm the bakery's reputation.

- **Disorganized Production Tracking and Status Blindness:** The kitchen staff does not have a centralized, real-time method to monitor where open orders stand within the preparation pipeline. Instead of using a shared visual board or tracking system, employees are forced to yell across the noisy kitchen room to ask about progress or constantly pause work to flip through stacks of loose paper receipts. This lack of visual order tracking creates operational confusion, as bakers and decorators must repeatedly stop their detailed, hands on tasks just to answer status inquiries from front counter workers who are trying to update waiting walk in clients or online messaging inquiries. These constant workplace interruptions disrupt focus, slow down overall production, and make it difficult to identify which urgent orders need to be prioritized.
- **Logistical Misalignments and Delivery Mix-ups:** Because pickup schedules, delivery addresses, and customer contact details are written down on loose paper slips, it is easy for kitchen staff to misread target dates or mix up address notes. Without a systematic coordination method to manage dispatch schedules, a baker might accidentally prepare and decorate a delicate custom cake days before the customer actually needs it. This causes the fresh product to sit in storage, losing its freshness before pickup. Conversely,

staff members might overlook a delivery slip entirely, realizing an order is due only when the customer arrives or sends a message, leading to severe fulfillment delays, emergency decor work, wrong delivery addresses, and logistics failures that damage customer trust.

- **Slow Hand-offs and Manual Record Keeping:** Confirming that an order was successfully prepared, paid for, and handed over to the buyer requires manual verbal follow-ups and physical file logging at the counter. If a tired worker forgets to file a completed order slip into the physical binder ledger book at the end of an exhausting shift, that entire sales record effectively vanishes from the day's bookkeeping. This missing data creates immediate financial blind spots and makes it impossible to verify if the order was properly completed or if cash balances match physical inventory.

3. Data Privacy and Bookkeeping Risks

- **Unsecured Business Records and Lack of User Privacy:** Customers lack a private interface where they can safely access their accounts and track their purchases, while business financial logs and customer contact lists are kept on loose paper notebooks or personal mobile phones. This setup leaves sensitive customer data exposed to physical theft, damage, or unauthorized eyes.

- **Slow Payment Tracking and Accounting Errors:** Managing down payments and cash balances using manual notebooks is labor intensive and prone to math mistakes. Because proof of payment screenshots from digital wallets are kept loose in mobile phone photo galleries, matching a payment screenshot to a specific paper order requires tedious manual cross checking, causing frequent cash discrepancy errors.
- **Absence of Business Insights and Data Blindness:** The bakery owner cannot easily check if the business is growing or calculate profit margins without spending hours doing manual math at the end of every month. To see how the bakery is doing, the owner must manually sort through and add up hundreds of individual physical order slips line by line, leaving management unable to track changing customer preferences or predict busy seasonal rushes.

1.3 Objectives of the Study

1.3.1 General Objective

The primary objective of this study is to design and develop a secure, web-based ordering, smart scheduling, and sales analytics platform for C&N Bakery to transition its manual transactional workflows into a unified online platform. The study seeks to replace paper-dependent habits—such as handwritten logbooks, physical kitchen wall calendars, and fragmented social media chat threads—with an integrated web platform that shares real-time data across customers, kitchen personnel, and store management. Ultimately, the system will eliminate pricing guesswork through automated cost computation, protect the kitchen from operational overbooking via automated scheduling guards, streamline order hand-offs using visual tracking pipelines, and provide visual sales insights to guide daily business decisions.

1.3.2 Specific Objectives

To successfully achieve the general objective of the study, the project will fulfill the following specific objectives:

1. System Analysis and Requirements Gathering:

To analyze and document the existing manual operational workflows of C&N Bakery regarding walk-in ordering, custom cake configuration, kitchen scheduling, and physical record-keeping, in order to establish the functional and technical requirements of the proposed platform.

2. System Architecture and Interface Design:

   To design the overall system architecture, database schema, and user interface wireframes, ensuring secure data handling, role based access control, and an intuitive user experience for both guest customers and bakery staff.

3. Storefront and Custom Cake Builder Development:

   To build a public facing web storefront featuring an interactive product catalog with category filtering and an automated custom cake builder that dynamically calculates itemized cost estimates in real time based on selected design parameters such as tiers, flavors, frostings, and size.

4. Structured Order Intake and Payment Submission:

   To implement a secure checkout process that enables logged-in customers to submit visual reference links, custom text specifications, delivery preferences, and proof-of-payment receipts directly into a centralized database.

5. Automated Kitchen Capacity Guard and Calendar Module:

   To develop an automated calendar scheduling module that tracks daily custom cake order limits and automatically restricts calendar date availability once maximum kitchen workload capacity is reached for a specific target date.

6. Kitchen Pipeline and Order Tracker Construction:

   To construct an administrative order management pipeline for kitchen staff that consolidates customer design choices into a single dashboard and provides a multi-stage status tracker to monitor production progress from preparation to final customer pickup.

7. Sales Analytics and Business Intelligence Dashboard:

   To create an administrative sales analytics dashboard that automatically aggregates historical transaction records and generates visual charts and reports to assist store management in tracking revenue trends and identifying top-selling products.

8. Authentication and Data Security Implementation:

   To integrate user authentication, password encryption, and input validation mechanisms to secure confidential financial records, safeguard customer personal information, and prevent data entry errors.

9. System Testing and Evaluation:

   To evaluate the performance, security, and usability of the developed web platform through functional software testing and standard user evaluation tools to verify its operational readiness for deployment

1.4 Scope and Delimitation of the Study

This study focuses on designing, building, and testing a secure web-based ordering, smart scheduling, and sales analytics platform specifically tailored to the daily operational workflows of C&N Bakery. The primary goal of the system is to transition the bakery’s manual, paper-dependent practices—such as handwritten order notebooks, physical kitchen wall calendars, and scattered social media chat conversations—into a single, unified digital platform. The software structure will be split into two primary components: a public customer portal for menu browsing and custom order submission, and a secure administrative dashboard to streamline kitchen operations, daily capacity limits, and business record-keeping.

1.4.1 System Environment and Accessibility

The platform will be developed exclusively as a responsive web application, eliminating the need for end-users or staff members to download, install, or update native mobile software. The user interface will utilize a flexible layout that automatically adapts screen components across various device display sizes, including desktop computers, laptops, tablets, and smartphones. The application will maintain cross-browser compatibility, operating reliably on standard modern web browsers such as Google Chrome, Mozilla Firefox, and Apple Safari. Access to the platform will require a device with an active internet connection to communicate with the cloud-hosted database server in real time.

1.4.2 User Roles and Access Controls

To keep store data accurate and secure, the system will divide user access into distinct roles:

- Bakery Customers: Public guest visitors will be able to freely navigate the platform to view the digital product menu, check item descriptions, and review store policies and disclaimers. Registered and authenticated buyers will gain access to a personal account dashboard where they can configure custom cake requests, submit order details, attach references, and track the progress of their active orders.
- Administrative and Staff Dashboard (Secure Backend Portal): This single backend interface will be shared by both the bakery owner and kitchen production staff upon authenticating their login credentials.
    - *Production and Kitchen View:* Kitchen staff and decorators will use this dashboard to review complete

order specifications, confirm production schedules, inspect customer-uploaded reference photos, update order progress through baking stages, and confirm order pickups or deliveries.

- *Administrative Controls:* The store owner will use this same backend portal to manage store-wide settings, adjust base menu pricing, toggle product visibility, set daily order capacity limits, and review automated visual sales analytics reports.

1.4.3 Scope of the Customer Portal

The customer-facing website will include all the operational features needed to guide a buyer from initial browsing to final order submission:

- Categorized Product Catalog: The storefront will organize items into clear product sections with category filtering options, helping users quickly find daily pastries, breads, and traditional baked delicacies.
- Live Price Calculator: This module will guide customers step-by-step through custom cake requests. As users select their design options such as tier count, dimensions, flavors, and frosting type a background script will

automatically compute and display the updated price estimate on the screen for total cost transparency.

- Design Reference Uploader: The order form will allow customers to attach visual inspiration links or upload reference images directly into the system, as well as type out personalized cake dedication text to prevent design and spelling mistakes.
- Order Limit Guard: During checkout, the customer will select a fulfillment date on an interactive calendar. Fulfillment dates that have reached the bakery's maximum daily kitchen workload capacity will be automatically disabled by backend scheduling rules. The form will then collect necessary pickup or delivery details.
- Self-Service Order Tracker: Logged in users will be able to monitor a status tracker inside their profile to see exactly what stage their cake is in as the kitchen team updates it, eliminating the need to call or text the shop for routine status updates.

1.4.4 Scope of the Administrative and Kitchen Dashboard

The secure backend dashboard will provide store management and the kitchen team with complete control over daily workflows, schedules, and transaction records:

- Bakery Performance Summary: Upon logging in, the owner will view an operational summary displaying key performance indicators, such as total revenue for the current month, pending customer tasks, and active kitchen workloads.
- Main Order Directory: This module will organize all incoming orders into a clean, searchable directory. Staff will be able to filter records by customer details, product types, pickup dates, and payment statuses without handling paper slips.
- All in One Design Layout: To avoid costly mistakes in the kitchen, opening an active order will present a single interface showing the customer's uploaded reference images alongside their selected flavor choices, size dimensions, and custom text notes.
- Step by step Production Status Controller: The kitchen view will allow bakers to update an order's status as it moves through different preparation stages . Staff will also be able to review and approve payment receipts submitted by customers.
- Instant Menu Visibility Toggles: Store staff will be able to turn off the visibility of specific menu items on the backend, which will remove out-of-stock pastries from the public storefront instantly during ingredient shortages.

- Automated Booking Limiter: This background feature will continuously count incoming orders for each date. The moment a daily capacity limit is met, the system will automatically block that date from accepting additional online reservations.
- Sales Analytics Visualizer: This administrative tool will gather historical transaction records from the database and generate clear visual charts, highlighting monthly revenue trends and the bakery's most popular product items.

1.4.5 Delimitation of the Study

To maintain a realistic project schedule and keep hosting costs manageable for a small enterprise, the following capabilities are explicitly excluded from the scope of this study:

- No Native Mobile Applications: The project will be strictly confined to a responsive web browser framework; it will not involve developing or publishing native mobile apps for Google Play or the Apple App Store.
- No Direct Automated Payment Gateway APIs: The platform will not connect directly to real-time banking APIs or automated payment processors. Payment verification will be managed manually by staff members who will inspect uploaded

proof-of-payment receipts and manually update payment statuses.

- No Third-Party Courier API Integrations: The application will operate independently of external delivery networks. It will not link directly with courier platforms (such as Grab or Lalamove); organizing delivery drivers and dispatch details will remain a manual task handled by store staff outside the system.
- No Raw Ingredient Inventory Tracking: The system will not calculate or monitor raw baking materials (such as the remaining weight of flour, sugar, butter, or eggs). Inventory awareness will rely on staff observation and the manual menu visibility toggles.
- No Payroll or Human Resource Tools: The platform is engineered solely for order taking, kitchen scheduling, and sales reporting. It will not include employee timecard tracking, staff shift scheduling, or payroll calculation modules.
- No Supply Chain or Supplier Logs: The system will lack features for managing ingredient suppliers, issuing wholesale purchase orders, or tracking fluctuating raw material costs from vendors.

- No Offline Operational Mode: The application will depend entirely on a live cloud database. It will not support offline data caching; users and staff will require an active internet connection to access the platform.

1.5 Significance of the Study

Building the Custom Cake Ordering Platform with Smart Scheduling and Sales Analytics for C&N Bakery will bring great practical value to everyone involved in the daily operations and growth of the establishment. As the local food and baking industry shifts toward digital spaces, traditional neighborhood bakeshops must update their tools to stay relevant and keep up with modern competitors. This study will serve as a practical demonstration of how a local bakery can successfully transition away from the disorganization of manual paperwork. By replacing easily lost handwritten logbooks, fading sticky notes, and scattered social media message threads with a centralized web application, this project will create a smooth, dependable ordering process. This system will ensure pricing transparency for customers, protect the kitchen team from accidental booking overlaps, and safeguard historical business records from physical damage or loss.

By modernizing these daily tasks, this study will provide clear, long term benefits to a wide group of stakeholders:

1. **Bakery Customers and Guests (The Buyers)**

Traditionally, buyers trying to order custom cakes or local delicacies face frustrating ordering roadblocks, including scattered chat histories, delayed responses, and unclear pricing structures. The platform will directly assist buyers by providing:

- **Honest and Instant Pricing:** Customers will benefit from complete price transparency, receiving real-time cost feedback as they select different cake configurations, removing pricing guesswork and confusion.
- **Accurate Design Submission:** The platform will allow buyers to seamlessly submit design reference photos alongside their flavor preferences and text specifications, minimizing design misunderstandings and ensuring expectations match the final product.
- **Self-Service Order Tracking:** Users will gain access to real-time status updates regarding active orders directly from their account dashboard, eliminating the need to wait for manual staff replies to confirm the current production phase.
- **Convenient Menu Browsing:** The centralized digital catalog will allow customers to comfortably browse offerings and plan their orders at any time from any internet-connected device.

2. Production and Kitchen Staff (Decorators and Bakers)

Kitchen staff working within a manual setup operate under daily pressure to track upcoming deadlines and unique design themes using fragile paper records. This project will introduce structure to the kitchen workspace by providing:

- **Workload Protection:** The platform will prevent accidental overbooking by regulating the daily volume of custom orders, protecting the kitchen team from exhausting workloads and operational burnout.
- **Production Accuracy:** Cake decorators will be able to review complete design instructions and reference images within a single, unified view, eliminating errors caused by searching through loose paper slips or disjointed chat threads.
- **Step-by-Step Production Tracking:** The kitchen interface will allow baking personnel to smoothly advance orders through verified preparation milestones, moving a cake from preparation to baking, packing, and final pickup.
- **Efficient Menu Management:** Staff will be able to update product availability instantly on the backend, preventing customers from ordering and paying for treats that are currently out of stock due to ingredient shortages.

3. Executive Management (The Bakery Owner)

For the business owner, managing a specialized shop without a centralized digital tool complicates financial record-keeping and business planning. This platform will streamline daily oversight by providing:

- **Automated Financial Records:** The system will organize transaction histories, down-payments, and payment verifications into a reliable digital ledger, reducing human calculation errors.
- **Visual Sales Insights:** The owner will gain access to automated performance tools that track long-term business growth and highlight top-selling products, facilitating data-driven inventory and financial planning.
- **Data Protection and Privacy:** Secure role-based access rules **will ensure** that sensitive financial metrics and customer details are restricted solely to authorized personnel, keeping business records safe from unauthorized eyes.
- **Strategic Business Decisions:** By providing clear digital summaries of customer buying trends, the platform will help the owner make smarter choices regarding ingredient budgeting and seasonal menu updates.

4. The Organization (C&N Bakery as a Business)

Transitioning to an automated platform will professionalize the C&N Bakery brand and protect its commercial reputation from the fallout of missed deadlines or forgotten details. The business will benefit from:

- **Prevention of Resource Waste:** Enforcing systematic confirmation and tracking parameters will ensure the kitchen never wastes costly ingredients on unverified or incorrectly noted custom orders.
- **Stronger Local Competitiveness:** The platform will grant a small, local shop a competitive edge by offering the modern digital conveniences that tech-savvy buyers expect today.
- **Scalable Business Growth:** By streamlining internal workflows and automating communications, the bakery will be able to safely handle higher order volumes during peak holiday seasons without overextending its administrative capabilities.

5. Future Systems Developers

For software engineers building e-commerce applications for specialized food markets, standard online stores lack the strict booking limits and custom configuration tools required for custom baking. This research will assist developers by offering:

- **Custom Order Reference:** This study will serve as a practical guide for building custom product forms that calculate prices in real time as users adjust design parameters.

- **Booking Rules Framework:** This study will provide a practical template for setting up automated calendar boundaries that prevent system overbooking based on fixed daily kitchen capacities.

6. Future Academic Researchers

This study will provide valuable documentation for researchers analyzing how small businesses transition from manual operations to localized web platforms. It will document real-world business workflows that can be directly compared to other retail models and will serve as a foundational reference for designing systems for made-to-order businesses. Furthermore, it will outline structured paths for future student projects, such as integrating automated SMS notifications or third-party logistics integrations.

7. The Local Food Service Industry

The success of this platform will demonstrate that practical, high-quality digital solutions are accessible to neighborhood bakeshops and small food vendors without requiring massive corporate software budgets. Raising local service standards through transparent pricing and digital reliability will encourage nearby food vendors to modernize their operations, elevating the industry as a whole.

8. BSIT Students

For Information Technology students, this documentation will serve as a practical blueprint for breaking down manual business problems into functional technical features. It will demonstrate how to translate real-world storefront operations into structured database states and systematic workflows, illustrating the core principles of effective user interface design and role-based data security.

1.6 Definition of Terms

To ensure a clear and accurate understanding of the technical concepts, system operations, and structural components discussed throughout this study, the following key terms are defined both conceptually (how they operate in general) and operationally (how they will be applied inside the C&N Bakery platform):

1. Web-Based System

- **Conceptual Definition:**

  A software application that runs on an online server and is accessed through a standard web browser over an internet connection, without needing local software installation on a user's device.

- **Operational Definition:**

  This refers to the custom online software platform engineered for C&N Bakery. It will allow customers, kitchen staff, and the bakery owner to view products, place custom cake pre-orders, track active orders, and view sales performance using any web browser on a phone, tablet, or computer.

2. User Authentication and Account Access Control

- **Conceptual Definition:**

A digital security system that checks a user's login credentials and verifies their account permissions before granting access to protected areas of a system.

- **Operational Definition:**

  This will serve as the system’s security checkpoint during login. Based on the user's account role, the system will automatically route buyers to the public ordering catalog, kitchen staff to production workflows, and the bakery owner to password-protected financial sales records.

3. Interactive Custom Cake Builder

- **Conceptual Definition:**

  A dynamic web interface that lets customers select, configure, and modify specific design options of a product before completing an order.

- **Operational Definition:**

  This will be the dedicated configuration page on the website where customers design a custom cake from scratch. Customers will select options like tier count, cake size, flavor, and frosting styles, while typing custom cake inscriptions and uploading visual design reference photos.

4. Automated Pricing Calculator

- **Conceptual Definition:**

A computer script running behind an interface that immediately calculates price totals whenever a user changes a selection on the screen.

- **Operational Definition:**

  This will be the calculation script operating inside the Interactive Custom Cake Builder page. The moment a customer selects a new option such as a larger cake size or a premium frosting the script will update the final price quote on the screen in real time, eliminating the need for manual price quotes over chat or phone calls.

5. Online Menu Catalog

- **Conceptual Definition:**

  An organized digital storefront that displays a business's product list in clear, searchable categories for easy browsing.

- **Operational Definition:**

  This will serve as the public digital display for C&N Bakery's product lineup. It will automatically organize items into categories—such as customized cakes, cupcakes, kakanin, and sapin-sapin—and will feature a search bar alongside price filters so shoppers can find treats that fit their budget.

6. Private Customer Dashboard and Progress Tracker

- **Conceptual Definition:**

  A private user account area that shows personal account details, past orders, and live status updates on active requests.

- **Operational Definition:**

  This will be the private account workspace for registered C&N Bakery customers. Users will be able to review their transaction history and track active orders through a visual status bar that updates as the kitchen marks an order as *Preparing* → *Baking* → *Packing* → *Delivery*.

7. Sales Analytics Charts

- **Conceptual Definition:**

  A data reporting tool that takes raw saved transaction records from a database and turns them into visual charts to show business trends.

- **Operational Definition:**

  This reporting tool will collect completed sales records and display them in visual tools: a *Monthly Revenue Graph* to track overall sales growth, and a *Top-Selling Products Graph* to highlight which products, like custom cakes or kakanin, bring in the most revenue.

8. Menu Availability Toggle Control

- **Conceptual Definition:**

  An administrative control switch that lets system managers quickly hide or show individual product listings in a public database catalog.

- **Operational Definition:**

  This will refer to the on/off availability switches next to each pastry item in the admin dashboard. If the kitchen runs out of raw ingredients for an item like sapin-sapin, staff can flip the switch to instantly hide that item from the public catalog, preventing customers from ordering sold-out treats.

9. Auto-Lock Booking Calendar

- **Conceptual Definition:**

  An automated scheduling rule that tracks daily reservations against a preset limit and automatically closes off calendar dates once that maximum threshold is reached.

- **Operational Definition:**

  This will function as the system's workload gatekeeper. It will continuously count incoming cake pre-orders for each date and automatically lock that day on the customer booking calendar as soon as it hits the bakery's daily limit, protecting kitchen staff from accidental overbooking.

CHAPTER 2

2.1 Foreign Studies

Mobile e-commerce platforms and cloud backends play a central role in simplifying everyday tasks for small, independent food businesses. Liang et al. (2021), in *A Mobile E-Commerce System for Local Bakery Management using Flutter and Firebase* (IEEE Xplore, ISSN: 2169-3536), examined how mobile software and cloud database tools help local bakeries manage their daily operations. The authors observed that small bakeries frequently run into trouble when they rely on manual paper logs, paper slips, or unorganized social media chat threads to write down incoming customer orders. When sales volume increases during busy weekends or seasonal holidays, paper order slips easily get misplaced, misread, or damaged near kitchen prep tables, which invariably leads to forgotten cake orders, wrong delivery times, and confused baking staff.

To solve this persistent operational problem, the researchers built a cross-platform application that allowed customers to browse bakery catalogs, select items, and place orders through their smartphones or web browsers. They connected the frontend user application directly to a real-time cloud database system. Whenever a customer submitted an order online, the web application instantly updated the store manager's control panel without requiring anyone to manually refresh the browser page. The evaluation revealed that real-time database syncing successfully eliminated duplicate order records, cut down order processing delays, and kept store staff instantly informed about newly incoming purchases.

This study is highly relevant to the proposed platform because it proves that real-time database tools keep customer orders and kitchen preparation schedules synchronized without human transcription errors. Traditional manual logs create unnecessary delays and risks for local bakeshops, whereas a synchronized cloud database ensures that custom order details are preserved accurately. The proposed project adapts a similar real-time database architecture to ensure that whenever a customer places an order or customizes a cake online, the bakery manager's administrative dashboard updates immediately, preventing lost orders and missed pickup deadlines.

Interactive customization tools are also essential when a food business sells made-to-order goods like specialized event cakes. Rajaput et al. (2022), in *Bakery Shop Cake Customization Management System* (Genesis Global Publication, ISSN: 2582-7421), investigated how interactive ordering systems reduce miscommunication between cake buyers and bakers. The authors pointed out that ordering a custom cake is far more complicated than buying pre-baked, standard items off a retail display shelf. Customers usually want specific cake dimensions, multi-tier structures, unique flavor combinations, custom frosting colors, specific decorative toppings, and short text inscriptions written on top of the cake. When these intricate details are discussed over simple phone calls or open-ended messaging apps, misunderstandings happen frequently, leading to disappointed customers and wasted baking ingredients.

To address this issue, the researchers designed an interactive web builder that guides customers through a step-by-step customization process. Instead of typing long, unorganized text messages, buyers select options from organized visual menus, such as choosing the number of cake tiers, selecting base sponge flavors, picking frosting styles, and uploading reference images for the baker to inspect. The testing results confirmed that offering structured choices through a dedicated web interface significantly improved overall order accuracy, saved administrative time for store staff, and reduced customer complaints regarding incorrect cake designs.

This foreign study provides a practical visual blueprint for the proposed platform’s custom cake module. By offering clear drop-down choices, selection cards, and upload input fields directly on the website, the proposed platform will help customers specify their exact cake design upfront without needing back-and-forth messaging. This structured approach eliminates guesswork for the local baker, ensuring that the final custom cake matches the buyer's exact design specifications while saving time for both parties during the ordering process.

Web system architecture must also accommodate the clear operational differences between ready-made baked goods and custom orders. Mali (2020), in *Fast and Efficient Cake Shop Management System Using Web-Based Architecture* (Technoscience Academy, ISSN: 2395-6011), studied how structured web applications streamline daily management tasks for small cake shops. The author explained that standard bakery inventory, such as pre-baked breads or display-case pastries, follows a simple sales process where stock counts are subtracted immediately upon purchase. On the other hand, custom cakes require dedicated labor planning, raw material preparation scheduling, and multi-stage order status tracking over several days.

When a store manager tries to track both standard off-the-shelf sales and custom order schedules in the exact same physical logbook, daily bookkeeping becomes messy, time-consuming, and confusing. The researcher built a web architecture that separated ready-made retail sales from custom order workflows into two distinct administrative dashboard views. The findings confirmed that separating these two product pipelines reduced administrative stress for business owners, simplified daily transaction records, and improved overall operational response times during peak business hours.

This study offers useful architectural guidance for the backend layout of the proposed system. Custom cakes demand a completely different preparation and tracking workflow compared to everyday baked goods, making separate management views necessary for efficient store operations. Mali's research justifies dividing the proposed platform's administrative panel into separate modules for regular daily menu orders and custom cake preparations, allowing the local bakery owner to monitor immediate retail sales and upcoming custom baking schedules simultaneously without operational confusion.

Managing kitchen labor limits requires automated order scheduling to protect bakery staff from operational overload and burnout. Sharma et al. (2020), in *Smart Scheduling and Dynamic Capacity Allocation in Custom Order E-Commerce Systems* (Foundation of Computer Science, ISSN: 0975-8887), analyzed automated scheduling logic and capacity management for businesses that make hand-crafted or custom-built products. The authors explained that custom manufacturing relies heavily on human labor hours, which means every kitchen or workshop has a strict maximum daily output limit. In custom cake baking, taking on more detailed orders than the staff can physically decorate in a single day leads to rushed workmanship, late deliveries, compromised quality, and burned-out decorators.

To solve this widespread overbooking issue, the researchers developed a dynamic scheduling feature that calculated daily production limits based on available labor hours and order complexity. When total accepted orders reached the store's maximum threshold for a specific date, the web system automatically locked that calendar date on the online checkout page, preventing new customers from selecting or booking that day. This automated process protected store owners from accidentally taking on more work than their kitchen crew could physically handle during peak event seasons.

This study provides the primary theoretical and structural backing for the **Smart Scheduling** feature of the proposed platform. Custom cake baking is a labor-intensive process, and local bakeries must enforce strict daily limits to maintain quality and meet delivery deadlines. By implementing automatic calendar slot limits based on the concepts established by Sharma et al., the proposed system will protect local bakery staff from overbooking during busy dates and major holidays, ensuring that every accepted order is delivered on time with high quality.

Visual sales dashboards help small business owners track earnings without having to read through long, complicated numerical spreadsheets. Kumar et al. (2021), in *Real-Time Sales Analytics and Visualization Frameworks for Small-Scale Retail Businesses* (Elsevier Science, ISSN: 0164-1212), examined how automated data visualization tools help small enterprise owners make practical business decisions. The authors observed that small food enterprise owners rarely have the time or specialized analytical skills to review long line-item sales logs or extract numbers from manual accounting ledgers. As a result, many small shop managers make raw material purchases and menu pricing decisions based on rough guesswork rather than real financial data.

To solve this decision-making hurdle, the researchers developed an analytics framework that automatically collects raw database records and converts them into clean visual charts, such as monthly revenue bar graphs, daily sales volume trends, and pie charts showing popular product choices. The evaluation revealed that visual graphic displays allowed non-technical store owners to quickly spot top-selling menu items, recognize seasonal revenue surges, and plan raw material purchases well ahead of time.

This research directly informs the design and layout of the **Sales Analytics** module in the current project. Small bakery managers need clear, at-a-glance financial insights to manage raw ingredient costs and identify popular cake choices without spending hours on bookkeeping. By studying the visual chart layouts proposed by Kumar et al., the proposed system will provide local bakery managers with an intuitive, easy-to-read dashboard that automatically translates historical order data into clear visual reports, helping them make smarter business and purchasing decisions.

2.2Local Studies

Local information system development plans provide valuable insight into how Philippine bakeshops transition from traditional paper workflows to digital platforms. An information system plan for Manolette Bakeshop, documented in the Neliti Philippine Research Repository (Neliti Repository, ID: 492104), evaluated the operational challenges faced by established community bakeries in the Philippines. The study highlighted that many local bakeshops continue to rely on physical notebooks, paper receipts, and verbal communication between counter staff and kitchen workers. When handling custom cake requests, staff members had to write down specific design choices by hand, which frequently resulted in misplaced order slips, forgotten customer notes, and overall operational delays.

To solve these persistent administrative issues, the study outlined a structured digitalization roadmap focused on centralizing order tracking, digitizing customer records, and establishing clear operational workflows. The roadmap provided a step-by-step model for replacing manual order logs with an interconnected digital system that connects counter staff, bakery owners, and kitchen decorators. The results of the plan showed that adopting digitized management tools reduced human record-keeping errors, eliminated lost receipts, and streamlined daily communication across departments.

This local study is directly relevant to the proposed platform because it proves that traditional Philippine bakeries need structured digital management tools to replace error-prone paper workflows. Operating a custom cake business through manual notebooks leaves room for expensive mistakes and customer dissatisfaction. The proposed project adapts the core principles of this digitalization roadmap by providing a centralized web platform that stores every custom cake request in a digital cloud database, ensuring that custom order details are preserved accurately without physical paperwork.

Regional studies further emphasize the importance of online ordering and inventory integration for provincial bakeries. Santos et al. (2021), in *Development of an Online Ordering and Inventory System for Local Bakeshops in Laguna* (Philippine e-Journals, ISSN: 2094-1749), examined how web-based software helps small-scale food enterprises across Laguna modernize their sales process. The authors noted that provincial bakeshops struggle to balance raw material inventory with unpredictable daily walk-in sales and custom event orders. When ingredient tracking is handled manually, bakeries often run out of essential baking supplies—such as specialized cake flour, food colorings, or fondant—right before completing major custom orders.

To address these inventory bottlenecks, the researchers designed an integrated web application that connected an online ordering portal directly to an automated stock deduction panel. Every time a customer placed an order online, the system automatically checked available ingredient stock and updated inventory counts in real time. The study confirmed that combining online ordering with automated inventory tracking minimized stock shortages, reduced ingredient waste, and improved overall fulfillment speed for local bakery owners.

This research provides functional guidance for the proposed system by demonstrating how online ordering platforms must seamlessly sync with kitchen availability to keep operations running smoothly. Unplanned ingredient shortages ruin baking schedules and delay customer deliveries. By understanding the inventory challenges identified by Santos et al., the proposed platform will incorporate real-time order processing that helps local bakery managers monitor their daily production capacity and order volumes effectively, ensuring timely order fulfillment.

Web-based customer relationship management tools tailored for local micro-enterprises also play a crucial role in improving customer retention. Garcia et al. (2022), in *Web-Based Custom Cake Ordering and Customer Relationship Management System for Micro-Enterprises in Cavite* (Journal of Philippine Information Technology Research, ISSN: 2782-8883), investigated how specialized ordering software benefits home-based and micro-bakery businesses in Cavite. The authors pointed out that micro-bakery owners spend an excessive amount of time answering repetitive inquiries on social media apps regarding cake flavors, sizes, pricing estimates, and available pickup dates. This constant back-and-forth messaging takes time away from actual cake baking and decorating.

To solve this issue, the researchers developed a dedicated web platform equipped with an interactive custom cake builder and an integrated customer database. Customers could select their desired cake specifications online, calculate estimated costs automatically based on selected features, and track their order status without needing to text the owner directly. The study revealed that providing a self-service custom ordering portal reduced administrative workloads for micro-enterprise owners while significantly improving customer satisfaction rates.

This study serves as a strong functional model for the proposed project, reinforcing the value of providing a self-service online cake configurator that frees up the baker's time. Relying solely on social media chat threads causes severe operational bottlenecks for local Philippine cake decorators. By adopting the interactive ordering framework from Garcia et al., the proposed platform will allow local customers to select their preferred cake options directly on the website, reducing repetitive chat inquiries and allowing the baker to focus on cake preparation.

Workflow management and order status tracking are equally critical during the kitchen production phase. Aquino et al. (2020), in *Order Tracking and Workflow Management Platform for Community-Based Bakeries in Central Luzon* (Philippine Journal of Science and Technology, ISSN: 0115-2130), analyzed how structured order status tracking improves operational productivity in community bakeries across Central Luzon. The researchers observed that once a custom order is accepted, kitchen staff often lose track of which cakes need immediate preparation, leading to late pickups, rushed decorating, and customer dissatisfaction.

To resolve these production delays, the authors built a central workflow tracking system that categorized each order into distinct status stages, such as *Pending*, *In Progress*, *Decorating*, and *Ready for Pickup*. Kitchen staff updated the order status in real time through an admin tablet view, allowing customers to view the exact progress of their cake through a simple online tracking page. The evaluation demonstrated that clear status tracking improved kitchen efficiency, reduced customer inquiry calls, and ensured timely order fulfillments during busy operations.

This local study directly informs the order management module of the proposed platform, highlighting the necessity of step-by-step order tracking from kitchen preparation to final delivery. Without clear visual status updates, kitchen staff can easily mismanage prep priority. The proposed system will incorporate a structured workflow tracker based on the findings of Aquino et al., enabling the bakery manager to update prep stages seamlessly and keep customers informed about their cake's real-time progress.

The rapid shift toward digital payments among local micro, small, and medium enterprises provides necessary context for modern e-commerce systems. Soriano et al. (2023), in *E-Commerce Adoption and Digital Payment Integration Among Small Food Enterprises in Metro Manila* (Research Congress on Philippine Business and Information Technology, ISSN: 2546-0617), evaluated the adoption rate of digital payment channels among small food vendors and bakeries in Metro Manila. The authors explained that local consumers increasingly expect small food businesses to offer cashless payment methods, such as GCash and Maya, alongside traditional cash transactions.

The researchers analyzed businesses that integrated automated e-wallet payment options into their ordering platforms against those that relied solely on manual bank transfers or cash on delivery. Their findings confirmed that integrating digital payment methods significantly increased completed checkout rates, reduced unpaid reservation cancellations, and simplified daily cash reconciliation for business owners. Offering instant digital payment channels gave customers greater confidence and convenience during checkout.

This study provides direct justification for integrating e-wallet payment verification into the proposed platform. Custom cake orders require deposit payments before baking begins to avoid wasted ingredients from bogus buyers. By building upon the digital payment insights of Soriano et al., the proposed system will include convenient e-wallet payment verification, ensuring that local cake buyers have a seamless, secure, and convenient checkout experience while protecting the bakery from bogus reservations.

2.3 RELATED LITERATURE

Academic literature on local software frameworks highlights the structural requirements necessary for building automated bakery order platforms. Perez and De Leon (2022), in *Frameworks for Automated Web-Based Ordering and Operational Management in Small Food Enterprises* (Philippine Information Technology Education Journal, ISSN: 2423-1126), evaluated the structural software standards required for specialized food ordering applications. The authors emphasized that generic e-commerce templates fail to address the specific operational constraints of local bakeries. Standard online shopping carts are designed for pre-packaged, ready-to-ship items and lack built-in logic for tracking custom cake specifications, calculating dynamic prep times, or managing daily baking limits.

Their literature review established that integrating custom order configuration, real-time schedule locking, and visual revenue reporting into a single web application creates a comprehensive solution for small-scale food enterprises. The authors outlined functional specifications for multi-tier selection tools, administrative schedule management, and automated sales graphing. Their framework demonstrated that combining these operational modules under one system improves order management efficiency while providing small shop owners with essential business control.

This literature serves as a primary structural benchmark for the current project, establishing the exact core modules needed to address local operational challenges. By adopting the software framework established by Perez and De Leon, the proposed platform will integrate a custom cake builder, a smart scheduling calendar, and a sales analytics dashboard into a unified web application. This ensures that the platform addresses the specific operational needs of local Philippine bakeshops rather than relying on inadequate generic e-commerce templates.

National policy reports and economic frameworks emphasize the urgent need for digital adoption among Philippine micro, small, and medium enterprises. The World Bank Group and the Department of Trade and Industry (2022), in *Digitalization and E-Commerce Adoption Among Micro, Small, and Medium Enterprises (MSMEs) in the Philippines* (World Bank Group & DTI Joint Policy Report, ISBN: 978-971-9123-45-6), evaluated the digital readiness of small businesses across the country. Their publication noted that over ninety percent of registered food enterprises in the Philippines operate as MSMEs, with many still relying on manual bookkeeping and traditional face-to-face transactions.

The report highlighted that micro-enterprises adopting digital channels experience higher operational efficiency, improved market reach, and greater resilience against economic disruptions. However, the report also identified major barriers to technology adoption, including the complexity of existing software platforms and high subscription costs for small shop owners. The authors advocated for simple, accessible, and localized web platforms tailored to specific enterprise needs, ensuring that small business owners can operate them without expensive technical setup costs.

This official literature provides macroeconomic justification for the current project, proving that digital transformation is essential for local bakeries to maintain profitability and competitiveness in the modern economy. Small food businesses cannot thrive on manual methods alone in an increasingly digital market. By following the policy recommendations of the World Bank Group and DTI, the proposed platform will offer a simple, accessible, and localized web solution that helps community bakeries transition to digital e-commerce without facing steep software learning curves or high financial barriers.

Regulatory frameworks from central financial institutions further underscore the importance of integrating modern e-wallet solutions into local e-commerce systems. The Bangko Sentral ng

Pilipinas (2023), in *Digital Payments Transformation Roadmap and the Growth of E-Wallets in Small Retail Operations* (BSP Policy Framework Series, ISSN: 2719-0129), outlined the national initiative to transition Philippine retail commerce toward a cash-lite economy. The BSP documented a rapid increase in consumer reliance on digital wallets like GCash and Maya, noting that local shoppers now prefer electronic payment options for daily purchases and online food orders.

The framework emphasized that small food businesses incorporating direct e-wallet payment options experience fewer cancelled reservations, faster payment settlements, and simplified financial tracking. Relying exclusively on cash on delivery or manual bank transfers creates transaction friction and financial risk for micro-enterprises. The report stressed that modern web applications must accommodate local payment gateways to meet changing consumer habits, streamline financial audits, and support national financial inclusion goals.

This literature provides direct regulatory backing for the proposed system's payment integration feature. Custom cake orders require advance deposits to guarantee order commitments and prevent financial losses from uncollected items. By aligning with the national digital payments roadmap established by the Bangko Sentral ng Pilipinas, the proposed platform will integrate convenient e-wallet verification options, giving local customers a familiar checkout method while ensuring secure reservation deposits for the bakery owner.

Technical literature on modern database design explains the necessity of cloud synchronization in interactive web applications. Rogers et al. (2021), in *Architectural Principles of Cloud-Based NoSQL Databases and Real-Time Synchronization in Web Applications* (ACM Digital Library / Academic Press, ISBN: 978-1-4503-8000-3), detailed the structural advantages of using cloud-hosted NoSQL databases for dynamic e-commerce platforms. The authors explained that traditional relational databases often experience performance lag or indexing delays when handling rapid, concurrent real-time updates from multiple simultaneous users.

In contrast, non-relational document databases allow continuous, multi-user data synchronization without server slowdowns or system crashes. When a customer modifies a custom order specification or books a delivery date on the frontend web interface, the database instantly broadcasts that update to the admin dashboard. The research confirmed that real-time document-based syncing maintains data integrity, reduces server load, and delivers an instant response across multi-device administrative setups.

This literature provides essential technical justification for selecting a real-time cloud database architecture for the proposed system. A cake ordering platform requires constant data exchange between frontend customer inputs and backend kitchen dashboards. Applying the cloud architectural principles outlined by Rogers et al. ensures that customer orders, custom cake selections, and schedule slot locks update instantaneously across the proposed platform without requiring manual web browser page reloads.

Human-computer interaction guidelines provide clear visual standards for designing intuitive product configurators in e-commerce applications. Martin et al. (2020), in *Human-Computer Interaction Guidelines for Custom Product Configurator Interfaces in E-Commerce* (IEEE Transactions on Human-Machine Systems, ISSN: 2168-2291), examined interface usability for customized ordering platforms. The authors explained that non-technical buyers experience decision fatigue and confusion when confronted with overly complicated, unstructured order forms filled with excessive text boxes and confusing choices.

To optimize user interaction, the authors recommended implementing step-by-step wizard interfaces, clear visual selection cards, dynamic cost updates, and explicit input fields. Their research proved that intuitive configurator designs increase order completion rates, reduce user input errors, and significantly improve overall user satisfaction. Structuring custom choices logically helps users complete complex orders quickly without feeling overwhelmed.

This technical literature directly informs the visual layout of the proposed system's cake customization page. Ordering a custom cake involves selecting multiple variables, such as size, flavors, frosting styles, and decorations. By applying the human-computer interaction principles recommended by Martin et al., the proposed platform will feature a clean, user-friendly cake builder that guides local customers through the customization process effortlessly, ensuring an enjoyable user experience while capturing accurate order details.

2.4 SYNTHESIS OF THE STUDY

The literature and studies reviewed in this chapter collectively emphasize the crucial role of digital transformation, specialized custom ordering workflows, dynamic calendar scheduling, and visual sales analytics for small food and retail enterprises. Both foreign and local researchers agree that traditional, paper-based record-keeping or unorganized social media messaging methods create severe operational bottlenecks for growing bakeries. Relying on physical notebooks, verbal kitchen handoffs, or static messaging threads frequently leads to lost order slips, delayed production, misplaced customer preferences, and significant administrative fatigue for small bakery owners.

Foreign studies demonstrated the technical feasibility and efficiency of modern web architectures in eliminating these manual administrative bottlenecks. Liang et al. (2021) and Mali (2020) proved that integrating real-time cloud databases and separating off-the-shelf inventory from custom order pipelines drastically reduces administrative processing delays and prevents duplicate order entries. Furthermore, Rajaput et al. (2022) established that interactive, step-by-step cake customization builders eliminate buyer-baker miscommunication by capturing design specifications upfront, while Sharma et al. (2020) highlighted the necessity of dynamic capacity allocation algorithms to prevent kitchen staff burnout caused by order overbooking. In terms of business management, Kumar et al. (2021) showed that visual sales analytics dashboards allow non-technical store owners to quickly interpret revenue trends and make informed inventory purchasing decisions without sifting through complex accounting ledgers.

Local studies provided essential context regarding the specific operational realities, structural constraints, and digital readiness of bakeries across the Philippines. Research from the Neliti Repository (2019) on Manolette Bakeshop, along with studies by Santos et al. (2021) in Laguna and Garcia et al. (2022) in Cavite, confirmed that local MSME bakeries frequently struggle with misplaced paper order slips, ingredient shortages, and excessive manual messaging on social media apps. Furthermore, Aquino et al. (2020) demonstrated that structured, step-by-step order status tracking improves kitchen productivity in Central Luzon bakeries, while Soriano et al. (2023) highlighted the rapid adoption and necessity of integrating e-wallet payment options like GCash and Maya to secure reservation deposits for food enterprises in Metro Manila.

The related literature reinforced these empirical findings from institutional, regulatory, technical, and human-computer interaction perspectives. Academic literature by Perez and De Leon (2022) established that standard off-the-shelf e-commerce templates are inadequate for custom cake operations because they lack specialized customization logic and dynamic schedule-locking capabilities. Policy reports from the World Bank Group and DTI (2022), alongside national digital payment frameworks from the Bangko Sentral ng Pilipinas (2023), provided macroeconomic and regulatory justification for digitalizing local food enterprises through accessible e-commerce and cashless payment channels. Finally, technical literature by Rogers et al. (2021) on cloud NoSQL database synchronization and design guidelines by Martin et al. (2020) on human-computer interaction supplied the theoretical foundation for building an intuitive, real-time web application tailored to small business operations.

The Research Gap

Despite the extensive research on bakery management, e-commerce adoption, and custom order platforms, a significant operational gap remains for local community bakeries in the Philippines. Existing foreign and local systems usually address these features in isolation—offering either a standalone cake builder, a basic inventory tool, or a generic appointment booking calendar. Most commercially available platforms are either too expensive for local micro-enterprises, overly complicated to set up, or lack dynamic scheduling logic that automatically locks calendar dates based on daily kitchen labor capacity. Furthermore, generic ordering templates rarely combine custom cake configuration, dynamic slot blocking, local e-wallet payment verification, and visual sales analytics into a single, unified web platform tailored to local Philippine bakeshop operations.

Additionally, most established e-commerce solutions cater primarily to large-scale retail franchises with standardized product lines. Micro and small bakeries in provincial communities operate under unique labor constraints, where a single master baker or small kitchen team must balance daily walk-in bakery sales with highly detailed, labor-intensive custom cake orders. When existing platforms fail to account for daily kitchen labor thresholds or lack localized payment verification methods like GCash, local bakers are forced back into manual social media messaging threads, re-exposing their operations to lost orders, overbooking, and uncollected reservations.

Contribution of the Proposed Study

The proposed platform addresses this research gap by integrating these essential operational tools into one accessible, web-based system designed specifically for local Philippine bakeshops. By synthesizing the architectural models of Liang et al. (2021) and Mali (2020), the proposed system will utilize a real-time cloud database to keep customer orders and kitchen management panels synchronized instantly. Adopting the step-by-step customization approach of Rajaput et al. (2022) and the user interface guidelines of Martin et al. (2020), the platform will feature an intuitive online cake builder that captures complete order specifications—such as tiers, flavors, frosting styles, and reference image uploads—upfront during checkout.

To solve the issue of operational overbooking highlighted by Sharma et al. (2020), the proposed system will implement smart scheduling logic that automatically calculates daily baking capacity and locks checkout slots when maximum limits are reached. Building upon the workflow tracking model of Aquino et al. (2020), the system will provide an administrative pipeline tracker so kitchen staff can manage order progress seamlessly from prep to pickup. Additionally, building upon the local findings of Soriano et al. (2023) and the BSP (2023) payment roadmap, the platform will integrate local e-wallet options to ensure smooth, cashless deposit transactions. Finally, drawing from the data visualization principles of Kumar et al. (2021) and the framework by Perez and De Leon (2022), the system will provide local bakery managers with an easy-to-read sales analytics dashboard to track income trends and top-selling products effortlessly.

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CHAPTER 3

This chapter presents the research design, system development methodology,system architecture, tools and technologies, data gathering techniques, testing plans, and evaluation metrics used to design, code, and deploy the web platform for C&N Bakery.

3.1 Research Design

In this study, we will use a descriptive-developmental research design. This simple, two-part approach combines two practical steps:

1. **The Descriptive Phase:** Observing, studying, and documenting how C&N Bakery manages its daily work using handwritten logbooks, wall calendars, loose paper slips, and physical receipts.
2. **The Developmental Phase:** Planning, coding, testing, and refining a web-based ordering and kitchen management platform that directly solves those daily struggles.

By connecting these two steps, we ensure that every button, database table, form input field, and dashboard tool we build addresses a real problem the bakery faces every single day, rather than making guesses about what they need.

PROPOSED DESCRIPTIVE-DEVELOPMENTAL RESEARCH METHODOLOGY

DESCRIPTIVE PHASE
• Observed walk-in counter orders and phone inquiries. • Documented problems with handwritten logbooks and paper slips. • Identified overbooking risks on the kitchen wall calendar. • Recorded manual math errors and bookkeeping fatigue at the counter.

DEVELOPMENTAL PHASE
• Will design the cloud database schema and user access roles. • Will build the Customer Portal & Custom Cake Builder page. • Will program automated daily order limits into the calendar. • Will build the Bakery Admin & Kitchen Order Dashboard.

3.1.1 Descriptive Research

Descriptive research focuses on observing and documenting how a business operates in real life. You do not change how the shop runs or set up fake tests in a laboratory. Instead, you look closely at everyday store routines to see where work gets slowed down and where errors usually happen.

Application in the Study

Before writing any code, our research team visited C&N Bakery to study their manual operations. We spent time inside the store, interviewed the business owner, talked with counter staff, and reviewed their physical order logbooks. Through these field observations, we identified four main operational problems:

FOUR PRIMARY STORE PROBLEMS
1. SCATTERED MESSAGES -> Orders spread across SMS, calls, and chat apps. 2. MISREAD PAPER SLIPS -> Illegible handwriting leads to cake design errors. 3. CALENDAR OVERBOOKING -> Wall calendar lacks automatic limits

for peak days.

4. MANUAL BOOKKEEPING -> Manual math on calculators causes errors.

1. Scattered Customer Messages and Notes

Customer inquiries and custom cake requests were spread across personal phone calls, regular text messages (SMS), and multiple social media messaging threads. When a customer contacted the bakery to check on an order or request a price quote, counter staff had to scroll through long, unorganized chat histories across different apps just to find reference photos, agreed price quotes, or requested pickup times.

- **Real-World Store Example:** A customer who inquired about a custom chocolate birthday cake on social media three days earlier calls the bakery to change the delivery time. Counter staff must stop helping walk-in customers to scroll through dozens of chat threads on a smartphone just to locate that specific customer's original message and cake design notes.

2. Lost or Misread Handwritten Order Slips

Specific custom cake choices—such as layer dimensions, flavor combinations, frosting types, custom color schemes, and written dedications—were written down by hand on loose paper slips. Writing details down by hand on loose paper created significant risks:

- **Illegible Handwriting:** Kitchen staff often struggled to read fast handwriting written during busy store hours, leading to mistakes in cake text spellings or wrong frosting colors.
- **Lost Paper Slips:** Loose paper notes were easily misplaced, dropped, or damaged near kitchen counters during high-volume prep shifts.
- **Real-World Store Example:** During a busy holiday weekend, an order slip for an 8 inch round vanilla cake with custom blue icing gets placed near the kitchen counter. Water spills on the paper, making the written text unreadable. Kitchen staff must pause prep work, search through text messages on a personal phone, or call the buyer directly to re-confirm the order details.

3. Booking Mistakes and Kitchen Overbooking

Pickup schedules and delivery dates relied entirely on a physical paper calendar notebook hanging on the wall. Because there was no automatic tracking system to count incoming custom orders, the bakery faced constant scheduling risks:

- **Kitchen Capacity Overload:** The kitchen team can safely bake and decorate a maximum of 3 to 5 complex custom cakes per day while maintaining quality standards.
- **Double-Booking Risks:** Counter staff taking phone orders during busy shop hours could easily forget to check the wall calendar before accepting a new custom order, causing accidental double-bookings and high operational stress during holiday seasons.
- **Real-World Store Example:** Counter staff accepts a phone booking for a 3-tier wedding cake for Saturday afternoon. Meanwhile, another staff member takes two custom birthday cake orders for the exact same Saturday without checking the wall calendar. On Saturday morning, the kitchen staff realizes they have double the work they can physically handle, leading to delayed deliveries and stressed bakers.

4. Tiring Manual Bookkeeping and Calculation Errors

Tracking customer down-payments, verifying online bank transfer receipts, and calculating daily sales totals required manual math using handheld calculators and physical logbooks. Doing manual math at the end of every long shift led to predictable operational struggles:

- **Mental Fatigue:** Calculating totals by hand after a long day of baking and customer service increased the chances of simple math errors.
- **Unverified Payments:** Staff members had to cross-check bank transfer receipts on a personal phone against physical paper logs, leading to long wait times at the counter.
- **Real-World Store Example:** At closing time, the owner manually adds up 40 physical paper receipts using a desktop calculator. If a single paper receipt is missing or a number is misread, the cash total does not match the paper ledger, forcing the owner to spend over an hour re-checking individual receipts.

3.1.2 Developmental Research

Developmental research is the step-by-step process of planning, coding, testing, and refining a software application over a structured project timeline. The main goal of this method is to take the real-world observations collected during the descriptive phase and turn them into a practical, working software tool.

It represents a hands-on building process where early versions of the application are coded, tested, reviewed by actual store users, and improved until the software runs smoothly under real-world conditions.

PUBLIC CUSTOMER PORTAL	ADMIN & KITCHEN DASHBOARD
* Menu with Budget Filters * Interactive Cake Builder * Automated Price Calculator * Live Self-Service Order Tracker	* Real-Time Summary Cards * Kitchen Order Stepper * Automated Calendar Rules * Visual Sales Graphs

Practical Application in Our Study

The developmental phase will form the practical building phase of this project. Using the observations gathered during store visits, the team will engineer a web platform featuring two connected portals linked to a single secure database:

1. The Public Customer Portal

An online storefront accessible from any smartphone, tablet, or desktop computer. It includes:

- **Organized Product Menu:** Clear categories (Cakes, Pastries, Custom Sets) with spending filters so customers can quickly find items within their budget.
- **Interactive Custom Cake Builder:** A visual selection tool where customers pick cake sizes, base flavors, frosting types, tier counts, and enter custom dedication text.
- **Automated Price Calculator:** A browser script that computes custom cake pricing instantly based on chosen options, removing the need for manual pricing calls.
- **Reference Image Link Box:** A dedicated text box where customers paste links to design inspiration photos, keeping visual references attached to the order.
- **Live Order Tracking Screen:** A self-service tracking view where customers enter their Order ID to see live progress without calling the store.

2. The Administrative and Kitchen Dashboard

A password-protected workspace designed for the bakery owner and kitchen staff. It includes:

- **Real-Time Summary Cards:** Clean screen displays showing pending orders, confirmed baking tasks, and daily pickup counts at a glance.
- **Order Directory & Design Inspector:** A structured list where staff can click any order to view exact cake specifications, customer notes, and reference links in one place.
- **4-Step Kitchen Order Stepper:** Simple status buttons that update progress across the entire system with a single click.
- **Visual Sales Graphs:** Automated income charts that summarize daily, weekly, and monthly sales instantly, eliminating manual bookkeeping math at closing time.

3. Automatic System Calendar Rules

Background logic built directly into the booking system to prevent overbooking:

- **Daily Capacity Counter:** The cloud database counts active custom cake orders for every date on the calendar.

- **Automated Calendar Locking:** When a specific date hits the kitchen's maximum capacity (such as 5 custom cakes), the checkout calendar automatically locks that date, preventing extra bookings and eliminating double-booking mistakes.

3.1.3 Why We Are Not Using Experimental Research

Experimental research is a laboratory method where researchers intentionally change one specific factor (such as an ingredient brand or oven heat) while strictly controlling all other factors across test groups to measure cause-and-effect results.

Plain-Language Explanation of Why Experimental Design Does Not Apply

REASONS EXPERIMENTAL RESEARCH WAS EXCLUDED
1. NO RECIPE TESTING -> We are building software, not testing baking tools. 2. APPLIED ENGINEERING -> Success means reliable code, speed,

and safety.

3. NO STORE DISRUPTION -> Forcing paper on test customers confuses buyers.

4. DATABASE FOCUS -> Focus is replacing paper notes with cloud storage.

1. We Are Building Software, Not Testing Baking Methods

This study will not conduct test batches in a kitchen to measure how changing ingredients or oven settings affects baking times, cake texture, or taste. The goal of this project is strictly to design, build, and deploy a web application that organizes daily orders and simplifies store management.

2. The Focus Is on Practical Software Engineering

This project focuses on software development, user interface design, and secure database setup. System success relies on practical engineering factors such as clean code execution, page load speeds, input validation security, and simple screen layouts for bakery staff rather than testing scientific formulas in a lab.

3. Avoiding Customer and Store Disruption

In a classic experimental study, researchers divide subjects into a "control group" and an "experimental group." Splitting real bakery customers into a control group forced to use slow paper notes and an experimental group using the new website would disrupt daily store operations, slow down counter service, and confuse buyers.

4. The Goal Is System Integration

Because the primary objective is replacing messy paper notes with a single, reliable online database, developmental research provides the exact right structure for creating user accounts, live price calculators, automated calendar locking rules, and secure database pipelines.

Table 3.1: Mapping Store Problems to System Methodology & Solutions

Observed Manual Problem	Descriptive Research Finding	Developmental Software Solution	Expected Operational Benefit
Scattered Messages	Notes spread across SMS, phone calls, and social	Single central order directory linked to	Staff view all order notes, reference links, and

	media chats.	customer accounts.	contact details on one clean screen.
Lost / Misread Notes	Handwritten paper slips get lost or damaged near kitchen counters.	Structured digital order forms with required text fields and options.	Clear, typed cake specifications prevent handwriting misinterpretations and lost notes.
Calendar Overbooking	The wall calendar lacks capacity checks, leading to double-bookings.	Automated calendar logic that counts active orders and locks full dates.	Eliminates accidental double-bookings and prevents kitchen burnout during holidays.
Tiring Bookkeeping	Manual math on calculators at closing time causes fatigue	Automated sales calculation dashboard and	Generates daily income totals instantly,

	and errors.	digital status logging.	saving time and eliminating math mistakes.

3.2 System Development Methodology

For this study, the research team will use the Agile Development Model as the primary software framework for the Software Development Life Cycle (SDLC). Traditional software methods, such as the classic Waterfall model, follow a strict sequence where one step must finish entirely before the next step can begin. Under Waterfall, making changes to earlier choices after coding starts is difficult and expensive.

Building a web application for a small business like C&N Bakery requires flexibility. As the bakery owner, kitchen staff, and customers begin using early versions of the site, practical operational needs will pop up that were not noticed during initial planning.

The Agile methodology solves this problem by focusing on flexibility, clear communication, and delivering working software features in short, manageable steps called sprints. Instead of waiting until the end of the project to test the system, the bakery owner and staff will test early module releases and give immediate feedback. This regular feedback loop will lower project risks, prevent the team from building unnecessary features, and ensure the platform fits smoothly into daily bakery routines.

7-PHASE AGILE SDLC METHODOLOGY FLOW
1. REQUIREMENTS -> Gather owner/staff needs and set scope boundary rules. 2. PLANNING -> Set up sprint schedules and cloud hosting limits. 3. DESIGN -> Draw page wireframes and design real-time database trees. 4. DEVELOPMENT -> Code system features across 3 focused building sprints. 5. TESTING -> Run unit tests, integration checks, and staff UAT sessions 6. DEPLOYMENT -> Launch live cloud servers and seed real bakery

data.

7. MAINTENANCE -> Monitor database speed, sync health, and staff logs.

1. Requirements Gathering

Definition:

This initial phase focused on identifying and listing the exact features, business rules, and performance standards the web application needed. The developers created a clear list of features, user steps, login security rules, and performance targets to keep the project on track and directly solve the bakery's daily operational problems.

Application in the Study:

The research team held preliminary discussions with the owner and staff of C&N Bakery to decide which features were necessary and which could be left out. To ensure the project finishes on schedule, complex corporate tools—such as live GPS delivery tracking and automatic inventory reordering—were excluded from the project scope. These discussions produced two basic lists of system requirements:

- Functional Requirements (What the system will do):

- A public product menu organized by categories and price ranges.
- An interactive Custom Cake Builder that calculates price estimates automatically.
- Input fields for custom cake inscriptions, size choices, and reference links.
- A user shopping cart and self-service order tracking page.
- A smart calendar booking engine that checks daily kitchen capacity in real time.
- A secure administrative order management dashboard with a 4-step progress tracker .
- Automated email updates sent to customers whenever their order status changes.
- Simple visual sales charts for daily, weekly, and monthly tracking.

- Performance Requirements (How the system will run):
  - Responsive Design: The web application will adjust cleanly across smartphones, tablets, and desktop screens.
  - User Access Control: Administrative and kitchen dashboards will be strictly protected behind encrypted password logins.

  - Speed and Efficiency: Public and administrative pages will load quickly, even on basic mobile data connections.

2. Planning

Definition:

The planning phase breaks down system requirements into manageable coding tasks spread across structured development sprints. It establishes the technical tools, sets up the cloud database plan, and ensures everything stays within project budget and database storage limits.

Application in the Study:

The development team will plan the website using standard web tools connected to Google Firebase services. To keep operating costs at zero and stay strictly within Firebase's free storage tier, the team established two practical operational rules:

- Customers will submit cake design ideas by pasting web links during checkout rather than uploading large, heavy image files directly to the database.

- Proof of payment receipts will be sent directly to C&N Bakery’s official Facebook Messenger page so staff can check them manually.

3. Design

Definition:

The design phase converts feature lists into visual page layouts and database structures before any coding begins. Creating these visual blueprints early helps catch layout errors on paper, saving valuable development time later.

Application in the Study:

Visual layouts and wireframes will be sketched for both the customer-facing storefront and the staff admin dashboard. Database structures will then be set up inside the Firebase Realtime Database to keep records organized:

- Order Data Structure: Cake options, design notes, reference links, pickup dates, and payment statuses will link directly to a single, unique Order ID so records do not get mixed up.

- Smart Calendar Rules: Database paths will be configured to count active custom orders for any chosen date. This allows the system to compare incoming bookings against daily limits and lock out full dates during checkout automatically.
- Sales Data Structure: Completed transactions will be stored with clear status labels, enabling the admin dashboard to calculate daily income totals quickly without lagging.

4.Development

Definition:

This is the phase where active coding occurs. Developers write the code, build the web screens, connect cloud backend services, and turn design plans into a working web platform. In Agile, development is split into short, focused coding sprints.

Application in the Study:

Development will be carried out across three structured sprints:

- Sprint 1 (Customer Portal): Will focus on building public storefront pages, adding login checks before accessing the Cake Builder, writing the automatic pricing script, creating checkout forms, and setting up the order tracking screen.

- Sprint 2 (Admin Management Portal): Will focus on building secure staff login screens, the Schedule Management Module, the Menu Management Module, the Orders Directory, and the Order History Module.
- Sprint 3 (System Integration): Will focus on connecting the user screens to Firebase cloud services, ensuring that status changes made by admins update live on the customer's tracking screen, and setting up automatic email alerts.

5.Testing

Definition:

The testing phase verifies that the website functions correctly, remains secure, and runs smoothly across different devices before launching to real users. In an Agile setup, testing happens continuously after each development sprint to catch bugs early rather than waiting until the end of the project.

Application in the Study:

The research team will run a multi-step testing plan across each development cycle. This includes four core testing activities:

1. Unit Testing: Checking small individual functions (like price calculation formulas and form input validation) on their own.
2. Integration Testing: Ensuring data transfers smoothly between web forms, user accounts, and the Firebase cloud database.
3. End-to-End System Testing: Running complete practice orders from start to finish—from custom cake design and calendar selection to order updates and sales reporting.
4. User Acceptance Testing (UAT): Holding hands-on trial sessions with key C&N Bakery users (the business owner, bakers, counter staff, and test customers) to confirm the site is clear and easy to use.

Note: To keep the thesis clear and avoid repeating information, the exact test cases, testing conditions, target results, and UAT feedback plans for these four stages are detailed fully in Section 3.6 (Testing Procedures).

6. Deployment

Definition:

Deployment involves uploading the tested web application onto live cloud hosting servers, configuring backend database nodes, and preparing the system for daily bakery operations.

Application in the Study:

The website will be hosted on cloud servers connected to Google Firebase backend services. Before opening the site to real customers, the development team will clean the database to remove all test accounts, fake orders, and sample records. The fresh database will then be populated with C&N Bakery's actual menu items, official prices, item categories, and daily booking limits.

7. Maintenance

Definition:

In the Agile lifecycle, launching the system is followed by monitoring site performance, gathering feedback from daily users, and making minor fixes to keep the application stable over time.

Application in the Study:

A practical post launch maintenance routine will be followed:

- System Performance Monitoring: Database response times, real-time sync reliability, and calendar date-locking features will be checked regularly during live use.
- Feedback Log: A simple feedback notebook will be provided to the bakery owner and staff to record daily notes, operational observations, or minor issues during their first few weeks using the site. This will help ensure the system remains easy to use, secure, and helpful for daily business operations.

3.3 System Architecture

The web application will use a cloud-based architecture hosted on Google Firebase under its free usage tier. Unlike older web setups that require setting up and managing a local web server (such as Apache running PHP) and a separate offline database engine, this design will allow the front-end website to talk directly to cloud services over secure internet connections.

By using a cloud setup, the project will eliminate local server setup, complicated backend code maintenance, and monthly hosting fees. To make sure the system stays well within Firebase's free database limit of 1 GB, the architecture is designed to avoid storing heavy media files in the database. Instead of saving large picture files (like payment screenshots or custom cake reference photos) on cloud servers, customers will send visual proofs directly to C&N Bakery's official Facebook Messenger page. This keeps the main database light and fast, storing only basic text records like user accounts, cake options, order dates, status updates, and sales summaries.

3.3.1 The Client-Side Application (Front-End Layer)

Definition:

The client-side represents the visual front-end of the web application. It includes all the pages, buttons, forms, and menus that customers and staff see and click on inside their web browser on a phone, tablet, or computer.

Application in the Study:

The front-end screens for both the Customer Portal and the Admin Dashboard will be built using standard, reliable web technologies: HTML5 for page layouts, CSS3 for mobile-responsive styling, and JavaScript for user interactions. The client-side will handle key interactive features—such as the Custom Cake Builder, category filtering, and automatic total calculations right inside the user's web browser.

Running the pricing calculations directly in the browser will give customers instant price updates as they select cake sizes, flavors, and design options, without making them wait for a slow page reload.

3.3.2 Cloud Backend Services Layer

Definition:

Using cloud backend services allows a web application to connect directly to ready made online tools instead of building and running a physical server from scratch. The cloud provider handles background operations like user account checks, real time data saving, and server security.

Application in the Study:

Google Firebase will serve as the main cloud engine for the application, handling account access and data storage through two specific features:

Cloud Module	**Primary Role in the Application**
Firebase Authentication	Will manage customer account sign-ups, password logins, staff access levels, and admin security checks.
Firebase Realtime Database	Will store lightweight text records (accounts, menu items, order details, calendar schedules, and sales logs) and sync updates live across screens.

1. Firebase Authentication

- **Definition:** A cloud access service that handles user account sign-ups, login checks, password protection, and access permissions.
- **Application in the Study:** This module will act as the security guard for the website. It will handle customer sign-ups, account logins, and staff security rules. When a

user tries to open the Admin Dashboard or change menu items, Firebase Authentication will check their account level. If a regular customer tries to open restricted admin pages, the system will block them automatically, keeping bakery records safe.

2. Firebase Realtime Database

- **Definition:** A cloud database that organizes text information into structured trees and syncs changes live across all open screens without requiring manual page refreshes.
- **Application in the Study:** This module will serve as the main storage vault for the entire website. It will store text records for user accounts, product menus, cake options, active bookings, kitchen schedules, and income reports. Because it syncs data instantly, whenever an admin updates an order status on the 4-step progress tracker, that change will appear right away on the customer's order tracking page. It will also protect kitchen capacity by counting active bookings for any chosen date and automatically blocking out dates that hit the daily limit during checkout.

3.3.3 Step-by-Step System Data Movement

To clearly show how the system components are designed to work together, data will move through the proposed architecture following two planned operational workflows:

1. Placing a Custom Order (Customer Side):

    1. **User Input:** The customer will select cake sizes, flavors, and custom details on the Custom Cake Builder page.
    2. **Instant Price Calculation:** JavaScript running in the browser will calculate the total price estimate directly on the customer's screen.
    3. **Receipt & Image Submission:** The customer will paste web reference links into the checkout form and send their payment receipt directly to C&N Bakery's official Facebook Messenger page.
    4. **Data Storage:** The website will transmit the text details of the order (customer name, selected options, calculated total, pickup date, and initial **Pending** status) to the Firebase Realtime Database.
    5. **Calendar Lock Check:** The database will increment the active order count for that selected pickup date to ensure daily kitchen capacity limits are maintained.

2. Managing & Updating Orders (Admin Side):

1. **Admin Security Check:** The bakery owner or staff member will log into the Admin Dashboard through Firebase Authentication to verify their account clearance.
2. **Live Directory Display:** The dashboard will read active order text records from the Firebase Realtime Database and display them on screen without requiring a manual browser refresh.
3. **Status Update:** The staff member will click to update an order status (e.g., changing from **Confirmed** to **Baking**).
4. **Real-Time Synchronization:** Firebase will record the new status instantly, transmitting the updated milestone to the customer’s tracking interface and triggering an automated email notification.

3.4 Tools and Technologies Used

To build, host, test, and manage the web platform, the research team selected a practical stack of web technologies, development software, and cloud tools. The following sections describe each tool, providing its standard definition and detailing exactly how it will be used to develop the C&N Bakery platform.

Tech Stack Summary Table

Category	Selected Tool / Technology	System Role in the Project
Front-End Design	HTML5, CSS3, Vanilla JavaScript	Builds user screens, handles mobile layouts, and runs real-time price calculations.
Cloud Backend	Google Firebase (Auth & Realtime Database)	Handles secure user logins and syncs lightweight text records live across screens.

Cloud Hosting Tier	Firebase Spark Plan	Provides zero-cost cloud hosting within a 1 GB database storage limit.
Development Workspace	Visual Studio Code (VS Code)	Serves as the primary code editor for writing and organizing project files.
Version Control	Git & GitHub	Saves development milestones and maintains cloud backups of project source code.
Testing & Monitoring	Firebase Console & Chrome DevTools	Used to inspect real-time database records, test security rules, and check mobile screen layouts.

3.4.1 HyperText Markup Language (HTML5)

Definition:

HyperText Markup Language (HTML5) is the standard markup language used to structure content on the web. It uses structural tags—such as headings, text fields, navigation menus, buttons, and tables—to tell web browsers how to lay out page information.

Application in the Study:

HTML5 will serve as the structural framework for both the public Customer Portal and the private Admin Dashboard. The development team will use semantic HTML5 elements to build:

- **Custom Cake Builder Forms:** Entry fields, drop-down menus, text boxes for custom cake inscriptions, and link inputs for cake reference ideas.
- **Order Tracking Screens:** Clean layout cards and content frames that hold order details and progress status indicators.
- **Admin Control Panels:** Data tables, form layouts, and action buttons used by bakery staff to manage product listings, set daily booking caps, and review orders.

3.4.2 Cascading Style Sheets (CSS3)

Definition:

Cascading Style Sheets (CSS3) is a design language used to style the visual appearance of a website, including colors, fonts, spacing, and layouts across different screen sizes.

Application in the Study:

CSS3 will control the visual styling and mobile responsiveness across all pages. The development team will write custom CSS3 rules to achieve the following:

- **Mobile-Responsive Layouts:** Flexible grid and flexbox layouts will ensure that complex admin tables automatically rearrange into simple, stacked card views when viewed on smartphones.
- **Visual Status Indicators:** Color-coded styles will be applied to the 4-stage progress tracker to clearly distinguish between **Pending**, **Confirmed**, **Baking**, and **Ready for Pickup** orders.
- **Interactive UI Highlights:** Hover effects, active button styles, and live color feedback on pricing fields will make price changes clearly noticeable to users.

3.4.3 Vanilla JavaScript (JS)

Definition:

Vanilla JavaScript is the core web programming language used to create interactive webpage features, update page content dynamically, run math calculations in the browser, and communicate with cloud services without relying on heavy external frameworks.

Application in the Study:

Vanilla JavaScript will serve as the primary engine for client-side features and database interactions. The development team will write native JavaScript code to run three core features:

- **Live Price Calculation Script:** Mathematical scripts will calculate price estimates in real time as customers pick cake sizes, tier counts, flavors, and add-ons, updating the total price on screen instantly without reloading the page.
- **Date-Locking Calendar Engine:** Scripts will read existing order counts from the database, compare them against daily kitchen limits, and automatically block out fully booked dates on the checkout calendar.

- **Real-Time Cloud Communication:** Async routines will send user inputs to Firebase as structured text records and stream live order updates directly to the Admin Dashboard.

3.4.4 Firebase Authentication

Definition:

Firebase Authentication is a cloud security service that manages user account verification, password security, and access permissions.

Application in the Study:

Firebase Authentication will manage security for the website, removing the need to write custom password encryption or account management code from scratch. The research team will use this service to:

- **Manage Customer Accounts:** Handle customer sign-ups, email logins, password encryption, and active user sessions.
- **Enforce User Access Levels:** Verify staff credentials whenever someone tries to open restricted admin pages (such as financial sales reports or schedule settings), blocking unauthorized access automatically.

3.4.5 Firebase Realtime Database

Definition:

Firebase Realtime Database is a cloud-hosted database that organizes information into structured text trees and syncs changes live across connected devices without requiring manual page refreshes.

Application in the Study:

The development team will use this cloud database to store project records without managing physical database hardware. It will be used to:

- **Store Lightweight System Records:** Keep product catalogs, cake options, pricing rules, user profiles, active orders, kitchen schedules, and sales summaries neatly organized in JSON trees.
- **Enable Live Status Updates:** Instantly show new orders on the Admin Dashboard as soon as a customer checks out, and update the customer's order tracking screen live whenever staff change an order status.

3.4.6 Firebase Spark Plan (Free Hosting Tier)

Definition:

The Firebase Spark Plan is the zero-cost tier of Google's cloud platform, offering web hosting, authentication, and database services within set free-tier usage caps (including 1 GB of database storage).

Application in the Study:

The research team will use the Spark Plan to host the website and run backend services at zero cost. To stay safely within the 1 GB database storage limit, the system is designed to avoid storing heavy media files in the database:

- **Text-Only Storage**: The cloud database will store only lightweight text, numbers, dates, and JSON records.
- **Messenger Proof Submission**: Customers will send visual files (such as payment screenshots and cake design references) directly to C&N Bakery's official Facebook Page Messenger inbox, keeping the cloud database lightweight, fast, and completely free.

3.4.7 Visual Studio Code (VS Code)

Definition:

Visual Studio Code is a lightweight code editor equipped with syntax highlighting, automatic code formatting, and workspace management tools.

Application in the Study:

VS Code will serve as the primary workspace for writing and organizing system code throughout development. The research team will use VS Code to:

- **Write and Organize Code:** Draft and manage HTML5 markup, CSS3 stylesheets, and JavaScript files within a single clean workspace.
- **Maintain Code Quality**: Use built-in syntax checks to catch formatting mistakes, unclosed HTML tags, and code errors early during development.

3.4.8 Git and GitHub

Definition:

Git is a version control system used to track code changes and save history snapshots, while GitHub is an online code hosting platform that provides cloud backup and project file management.

Application in the Study:

Git and GitHub will handle version control and cloud backup for the project's source code. The development team will use these tools to:

- **Save Project Milestones:** Create clear progress snapshots (commits) after completing major web modules across Sprints 1, 2, and 3.
- **Protect Working Code:** Write and test new features on separate code branches before merging them into the main project code, keeping established features working smoothly.

3.4.9 Firebase Console Dashboard

Definition:

The Firebase Console Dashboard is a web-based management interface provided by Google that lets developers view cloud database records, monitor active account logins, adjust database security rules, and check network traffic.

Application in the Study:

The development team will use the Firebase Console to manage cloud backend operations during system setup and testing. The dashboard will be used to:

- **Configure Security Rules:** Set up and test database access rules to ensure that read and write permissions match user access levels.
- **Monitor Storage Limits:** Keep track of real-time database connections and storage usage to ensure the project stays well within Spark Plan free limits.
- **Inspect Live Data:** Review stored JSON nodes, verify that dynamic updates work during end-to-end testing, and fix backend configuration issues.

3.5 Data Gathering Techniques

To gather all the necessary business rules, daily operational needs, and system requirements for the web application, the research team used three qualitative data gathering methods: Interviews, Direct Observation, and Document Analysis. The main goal of this process was to closely examine C&N Bakery's manual, paper-based daily routines, find slowdowns in their workflow, learn their pricing formulas, and translate real-world business rules into exact software features.

DATA GATHERING METHODOLOGY
1. STRUCTURED INTERVIEWS -> Captured business rules, production capacities, and operational pain points from owner & staff 2. DIRECT OBSERVATION -> Tracked physical kitchen workflows, order states, and customer interactions during peak hours 3. DOCUMENT ANALYSIS -> Examined handwritten logbooks, paper receipts, pricing sheets, and historical product catalogs

3.5.1 Interviews

Definition:

An interview is a direct qualitative research technique where developers ask structured or open-ended questions to key business staff. This method allows researchers to gather detailed operational insights, understand real user needs, and define exact software features directly from the people running the business.

Application in the Study:

The research team held structured consultation sessions with C&N Bakery's key internal staff: the business owner (1), front-counter administrative staff (2), and head kitchen bakers (2). These sessions examined day-to-day business operations, focusing specifically on:

- **Pricing Rules:** How custom cake prices were calculated using cake sizes, flavor choices, frosting layers, and decorative add-ons.
- **Payment Bottlenecks:** The daily confusion caused by unconfirmed digital bank transfers and cash payments at pickup.

- **Kitchen Production Limits:** The maximum number of custom cakes the kitchen staff could safely bake and decorate in a single day without sacrificing quality.

Summary of Interview Insights to System Features

Key Stakeholder Insight	Observed Operational Issue	Planned Software Implementation
Kitchen Baking Limit	The kitchen could only bake 3 to 5 custom cakes per day without losing quality.	The backend database will track orders per date and automatically lock out full dates on the calendar.
Payment Confusion	Unconfirmed digital payments caused manual accounting delays at the counter.	Customers will submit payment receipts to C&N Bakery's Facebook Messenger page for quick staff checks.

Scattered Order Details	Custom cake notes got lost in personal SMS text threads during rush hours.	All inscriptions, design notes, and reference links will save under a single Order ID record.

Planned System Integration

- **Scheduling Capacity Rules:** Daily production limits provided by the owner will be converted directly into backend database logic. The system will maintain an active order counter for every calendar date, automatically locking checkout availability once a date hits its limit to prevent overbooking the kitchen.
- **Payment Verification Protocol:** To help counter staff verify payments easily while keeping cloud database costs at zero, the checkout process will instruct customers to send digital payment receipts directly to C&N Bakery's official Facebook Messenger inbox for quick manual verification.
- **Unified Order Customization Rules:** To prevent custom design details from getting lost in personal text messages or loose paper slips, the platform will capture all cake dimensions, custom text inscriptions, design notes, and web

reference links during checkout, saving them under one unique Order ID.

3.5.2 Observation

Definition:

Observation is a primary research technique where developers watch real-world activities as they happen in their normal work setting. This method allows researchers to record actual operational tasks, daily work habits, and time spent on processes without interrupting the business.

Application in the Study:

The research team spent time inside C&N Bakery during busy morning and weekend operational hours. The team watched how staff handled walk-in customers, wrote down paper orders, tracked baking progress in the kitchen, and calculated end-of-day sales totals.

Through direct observation, the team identified three primary operational slowdowns:

- **Task Visibility Gaps:** Kitchen bakers relied on verbal reminders or loose paper notes, making it hard to track which cakes were pending, currently baking, or ready for pickup.

- **Manual Calculation Delays:** Counter staff spent extra time looking up base prices on paper lists and calculating custom add-ons on handheld calculators while customers waited in line.
- **Manual Sales Summaries:** The business owner calculated daily earnings by hand-totaling stacks of physical paper receipts at the end of each business day.

Planned System Integration

- **Order Lifecycle Tracker (4-Stage Pipeline):** Watching how kitchen staff tracked baking tasks directly guided the layout of the Admin Dashboard. The system will feature a 4-stage visual progress tracker, allowing staff to update and monitor order statuses in real time.
- **Automated Sales Reporting Framework:** Observing how the owner compiled paper bills to calculate daily income guided the analytics features. The database will categorize all completed sales automatically, generating daily, weekly, and monthly sales summaries without manual receipt counting.
- **Categorized Menu Navigation:** Watching budget-conscious walk-in customers guided the storefront layout. The public website will include category filters and price-sorting

options so customers can quickly browse items that fit their spending budget.

3.5.3 Document Analysis

Definition:

Document analysis is a research procedure used to review physical or digital business records, forms, logbooks, and administrative files. Analyzing these real-world records helps developers understand established business rules, pricing data, and record-keeping habits.

Application in the Study:

The research team gathered and reviewed C&N Bakery's physical paper logbooks, handwritten order slips, official base price sheets, and product catalogs. Reviewing these physical records provided the exact math formulas, page structures, and business rules the bakery had used for years.

Document Analysis to Database Mapping

Physical Bakery Artifact	**Data Extracted From Document**	**Planned System Equivalent**
Handwritten Order Slips	Form fields for size, flavor, tiers, and custom text inscriptions.	Digital Custom Cake Builder forms and checkout input fields.
Printed Price Reference Sheets	Base prices per size and add-on costs for extra tiers/flavors.	Automated JavaScript price calculation engine running in the browser.
Highlighter-Marked Receipts	Colored markers showing paid deposits versus unpaid balances.	Automated database status tags (**Pending**, **Confirmed**, **Baking**, **Ready**).

Planned System Integration

- **Digital Form Architecture:** The physical layout of handwritten paper order forms—specifically fields for cake dimensions, tier counts, flavor choices, and icing types will serve as the blueprint for designing the digital Custom Cake Builder.
- **Mathematical Pricing Engine:** The team transcribed handwritten price sheets into exact math formulas within the front-end JavaScript code. This will ensure that any combination of cake size, base flavor, and decorative options generates an instant price estimate that matches C&N Bakery's official rates.
- **Structured Data Organization:** Analyzing past product catalogs guided the layout of the Firebase JSON database tree, ensuring menu items will be categorized logically into clear collections (such as Customized Cakes, Standard Cakes, and Pastries).
- **Standardized Status Tags:** Reviewing old paper receipts showed that staff used colored highlighters to separate paid orders from partial down-payments. To eliminate confusion, the system database will assign standardized status tags to every transaction, replacing physical highlighters with automated data fields.

3.6 Testing Procedures

To ensure that the C&N Bakery web platform functions accurately, processes customer orders safely, and runs smoothly during daily business hours, the research team will execute a structured, multi-stage testing plan. Testing will cover every core system module—from front-end price calculation scripts to real-time database synchronization between the customer storefront and the admin dashboard.

4 STAGES OF SYSTEM TESTING
1. UNIT TESTING -> Tests small individual parts like buttons, price calculators, search filters, and text boxes. 2. INTEGRATION TESTING -> Checks how different parts pass data to each other,like linking orders to Facebook Messenger. 3. SYSTEM TESTING -> Tests the full website flow from ordering a cake to track daily sales reports. 4. USER ACCEPTANCE (UAT) -> Tests the finished website with the bakery owner, kitchen staff, and real customers.

3.6.1 Unit Testing

Definition:

Unit testing involves examining individual code components or web interface elements—such as a single button, text input box, or price calculation script—in isolation. The primary goal is to verify that each small component functions properly on its own before connecting it to other system modules.

How It Will Be Conducted:

The development team will test small interface functions across three main operational areas:

- **Testing the Custom Cake Price Calculator:** The team will run math test scripts inside the cake builder by selecting various cake dimensions, tier counts, base flavors, and custom frostings. The calculated totals will be compared directly against C&N Bakery's official printed price list to confirm math accuracy.
- **Testing Menu Budget Filters:** The team will click through each price filter option to confirm that the storefront displays only products falling strictly within the selected price range.
- **Testing Form Validation Rules:** The team will test input forms by deliberately leaving required fields blank,

entering invalid email formats, or submitting incomplete phone numbers. This will verify that the system blocks improper submissions and displays clear error messages to the user.

Planned Unit Test Cases

Tested Component	**Test Condition / Input**	**Planned Test Objective**	**Success Criteria**
Cake Price Engine	8-inch round + 2 tiers + vanilla frosting + custom inscription	Verify that JavaScript calculates the correct total price automatically.	Calculated price matches the official bakery price sheet exactly.
Budget Filter	Selected price range: ₱300 – ₱500	Verify that product filtering hides	Displays only products priced

		out-of-budget catalog items.	between ₱300 and ₱500.
Form Validation	Submitted checkout form with an empty phone number box	Verify that input validation blocks incomplete form submissions.	Blocks form submission and prompts the user for a valid phone number.

3.6.2 Integration Testing

Definition:

Integration testing focuses on checking how different software modules pass data back and forth. This ensures that data transfers smoothly across web pages, user accounts, and backend cloud databases without errors or data loss.

How It Will Be Conducted:

The research team will test data communication between the front-end user interface, authentication services, and the Google Firebase Realtime Database:

- **Account Registration and Profile Syncing:** The team will test the link between customer sign-up forms and database records. When a new user registers, the system will create their account record, store their contact details, and link future orders to their personal profile history.
- **Real-Time Calendar Capacity Updating:** The team will test the connection between the checkout interface and the calendar database. Tests will verify that completing an order automatically updates the active booking counter for that selected date in real time.

- **Instant Order Status Synchronization:** The team will test live synchronization between the Admin Dashboard and the customer's tracking screen. When staff update an order status (e.g., changing from **Baking** to **Ready for Pickup**), the change will reflect instantly on the customer's display without requiring a manual page refresh.

Integration Testing Scope

Integration Focus	Testing Procedure	Expected System Behavior
Auth & Database Link	Register new user accounts and inspect stored Firebase user nodes.	User profile data saves securely and links properly to order records.
Checkout & Calendar Link	Place test orders on specific pickup dates	Daily order counts increase live and

	and monitor database counters.	lock out dates that hit daily limits.
Dashboard & Tracker Link	Change order status tags on the Admin Dashboard and watch the user screen.	Customer order status updates in real time without refreshing the page.

3.6.3 System Testing

Definition:

System testing evaluates the entire application from start to finish as a complete operational platform. This ensures that all end-to-end business workflows—from customer product selection down to admin sales logging—operate smoothly without system failure.

How It Will Be Conducted:

The research team will execute complete operational test runs to evaluate overall system stability:

- **Full Order Placement Workflow:** Team members will simulate customer ordering by choosing a cake design, picking an available pickup date, following instructions to send payment proof via Facebook Messenger, and checking if the order details record accurately on the Admin Dashboard.
- **Daily Booking Limit Enforcement:** To evaluate calendar locking features, the team will place sequential test orders on a single date until reaching the preset daily capacity limit. The team will verify that the system locks that calendar date automatically for subsequent checkout attempts.
- **Automated Sales Reporting Workflow:** Team members will progress test orders through all stages until marking them as **Completed.** The team will confirm that completed transaction values automatically accumulate in the admin sales summary dashboard.

End-to-End System Workflow Map

Phase	Customer Side Workflow	Bakery Admin Side Workflow
Step 1: Selection	Select cake dimensions, flavors, and design add-ons.	The system displays a live price estimate on the user screen.
Step 2: Scheduling	Pick an open date on the calendar builder.	Database checks daily booking limit for selected date.
Step 3: Verification	Submit order text and send receipt via Facebook Messenger.	New order appears on the Admin Dashboard instantly.

Step 4: Fulfillment	Monitor order progress on the live tracking page.	Staff verifies payment and updates status .
Step 5: Completion	Collect completed order at the bakery counter.	Staff marks order **Completed**; daily revenue totals update.

3.6.4 User Acceptance Testing (UAT)

Definition:

User Acceptance Testing (UAT) is the final evaluation phase where intended real-world users—including the bakery owner, counter staff, kitchen bakers, and bakery customers—evaluate the platform to confirm usability, screen clarity, and operational fit.

How It Will Be Conducted:

The research team will conduct structured evaluation sessions with representatives from all primary user groups:

- **Customer Portal Usability Evaluation:** A selection of bakery customers will complete realistic tasks, such as browsing the product catalog, customizing a cake, checking pricing, and tracking an order. The research team will observe user interactions to confirm that navigation and button placement are intuitive.

- **Administrative Operations Evaluation:** The bakery owner, counter workers, and kitchen staff will interact with the Admin Dashboard. They will practice reviewing incoming orders, verifying payment submissions, updating baking statuses, and adjusting product availability to ensure the interface supports daily store operations smoothly.

Planned UAT Participant Group Breakdown

Evaluator Category	**Participant Count**	**Primary Evaluation Tasks**
Bakery Owner	1	Reviewing sales summaries, setting store rules, and managing product listings.
Counter Staff	2	Reviewing incoming orders, checking payment receipts, and updating status tags.
Kitchen Bakers	2	Monitoring active baking schedules and tracking daily production volume.
Bakery Customers	10–15	Customizing cakes, filtering product menus, and tracking order progress live.

3.7 Evaluation Criteria

To measure the success, overall stability, and operational value of the C&N Bakery web platform, the research team established a structured evaluation framework based on five software quality criteria: Functionality, Usability, Reliability, Efficiency, and Security. This framework defines the specific testing rules and target performance metrics used to evaluate both backend cloud performance and the front-end user experience for customers and bakery staff.

5 SYSTEM EVALUATION CRITERIA

1. FUNCTIONALITY	Evaluates whether all features, calculations, and admin controls work completely and accurately as designed.
2. USABILITY	Evaluates how easy and comfortable the website is to navigate for both customers and bakery employees on any device.

3. RELIABILITY	Evaluates system stability, error recovery, and data safety during active user sessions and network flickers.
4. EFFICIENCY	Evaluates the speed of price calculations, live database updates, and automated daily sales reporting.
5.SECURITY	Evaluates account login rules, data privacy, and admin access protection to safeguard business records and user profiles.

Evaluation Rating Scale

During User Acceptance Testing (UAT), evaluators will rate system performance across each quality criterion using a standard 5-point Likert scale:

Rating Value	Evaluation Level	Quality Description

5	Strongly Agree / Excellent	Feature works perfectly without errors and exceeds daily operational expectations.
4	Agree / Very Good	Feature works smoothly with minor non-critical visual or operational adjustments needed.
3	Neutral / Satisfactory	Feature functions adequately but requires basic guidance or minor interface tweaks.
2	Disagree / Poor	Feature exhibits noticeable errors or layout confusion that hampers daily use.
1	Strongly Disagree / Very Poor	Feature fails to function, breaks system workflows, or generates incorrect outputs.

3.7.1 Functionality

Functionality measures whether the software accurately executes every feature, math calculation, and administrative control it was designed to perform.

Evaluation Protocol

The research team will establish test routines to check whether both the public storefront and the private admin dashboard execute their functions according to system specifications.

Planned Evaluation Metrics

- Customer Storefront Features: Evaluation will check whether the product catalog applies selected budget filters accurately and whether the Custom Cake Builder calculates live pricing correctly as users select different cake sizes, tier counts, and flavor options.
- Admin Dashboard Features: The protocol will verify that the admin control panel generates clear sales summaries and allows staff to move orders through each stage of the baking pipeline without display errors or dropped data.
- Order Notification Features: Evaluation will verify that order status changes trigger updates successfully on both customer tracking displays and administrative log screens.

Functionality Acceptance Target Matrix

Feature Evaluated	Target Test Condition	Expected Acceptance Criteria
Cake Price Calculator	Select 8-inch cake + 2 tiers + custom icing options	Calculated total price matches official bakery pricing sheet exactly.
Order Status Tracker	Staff moves order status tag from Baking to Ready for Pickup	Status tag updates live on both administrative and customer tracking views.
Catalog Budget Filter	Apply ₱300 - ₱500 price range filter	Display shows only products priced strictly between ₱300 and ₱500.

3.7.2 Usability

Usability measures how easy, clear, and straightforward the website is for customers and bakery employees to navigate on their own without requiring technical guidance.

Evaluation Protocol

The usability protocol will measure how easily and comfortably users complete daily tasks using smartphones, tablets, and desktop computers.

Planned Evaluation Metrics

- Customer Checkout Experience: Usability testing will evaluate whether customers can select a cake, enter custom design instructions, pick an available pickup date, and review payment submission directions on Facebook Messenger without experiencing confusion.
- Bakery Staff Interface: Evaluation will assess admin dashboard readability under active kitchen conditions. The protocol will verify that the layout adjusts automatically for mobile screens so counter staff and bakers can tap status buttons with one hand while managing kitchen tasks.

3.7.3 Reliability

Reliability checks how consistently the website operates over time without crashing, dropping order information, or freezing—even when internet connections fluctuate.

Evaluation Protocol

The research team will execute stress test routines to verify that the cloud database maintains operational stability and preserves order records during heavy web traffic or temporary network disruptions.

Planned Evaluation Metrics

- Simultaneous Order Handling: The evaluation protocol will submit multiple order requests at the same time to verify that the cloud database processes concurrent inputs smoothly without crashing or stalling.
- Data Preservation During Disconnections: The protocol will check whether user inputs and order selections save safely in local browser memory if a customer loses connection mid-checkout.
- Automatic Screen Reconnection: Evaluation will verify that the live order tracking view reconnects automatically once network service stabilizes, keeping admin screens updated without manual page refreshes.

3.7.4 Efficiency

Efficiency measures the speed and responsiveness of the platform, focusing on how quickly it calculates pricing estimates, updates database records, and automates daily tasks compared to traditional paper records.

Evaluation Protocol

The team will measure operational speed and response times using browser developer tools to verify how much faster the web system processes daily tasks compared to paper logbooks.

Efficiency Performance Benchmarks

Operational Task	Manual Paper Baseline	Target System Speed Metric	Verification Tool / Method
Calculating Cake Estimates	3 to 5 minutes (Manual	Instant (< 1 second) live calculation	Front-end JavaScript performance logging

	lookup and calculator)		
Kitchen Task Tracking	Verbal calls or loose paper slips	Instant (< 2 seconds) cloud database sync	Firebase Console real-time monitoring
Checking Date Availability	Manual paper logbook checks	Instant real-time calendar date locking	Chrome DevTools Network panel timing

3.7.5 Security

Security covers the access controls, user authentication rules, and data rules used to keep customer accounts, payment receipts, and administrative sales records safe from unauthorized access.

Evaluation Protocol

The research team will establish security checks to evaluate whether user accounts, business financial metrics, and customer contact information remain protected.

Planned Evaluation Metrics

- Admin Access Protection: The evaluation protocol will test database security rules to ensure that standard customer accounts are blocked automatically from accessing or editing administrative dashboard pages.
- Form Script Blocking: The protocol will test all form text inputs to confirm that unauthorized code scripts cannot be entered or run through public order fields.
- Private Business Data Safeguards: Security testing will verify that daily sales reports, total revenue numbers, and customer phone logs remain hidden from public view and are accessible only to authenticated administrator accounts.

CHAPTER 4

SYSTEM DEVELOPMENT, IMPLEMENTATION, TESTING, EVALUATION, RESULTS, AND DISCUSSION

4.2 Overview of the Developed System

The Custom Cake Ordering Platform is a web-based, mobile-friendly application built to simplify custom cake ordering, manage daily baking schedules, and track sales performance for C&N Bakery. The system helps business owners and staff organize custom cake details, set daily order limits, track baking progress, and monitor sales trends. By replacing manual paper orders and scattered messaging chat records with a digital database and visual reports, the application helps the business make better, data-backed decisions.

Unlike traditional Point-of-Sale (POS) systems that only record physical counter sales, this platform includes an automatic price calculator, a calendar booking control system, and sales analytics reports. The application processes chosen customer options—such as tier counts, layer sizes, flavors, and design details—to calculate order prices instantly. It also uses past sales records to generate simple dashboards that show sales totals, performance charts, and popular menu items, giving the business owner clear insights into daily operations. Overall, the platform offers C&N Bakery an easy-to-use solution that improves kitchen workflow, prevents order overbooking, helps with ingredient planning, and securely stores customer records.

The major objectives of the developed system are as follows:

- Automate the recording, price calculation, and storage of custom cake orders.
- Prevent kitchen overbooking by automatically locking fully booked calendar dates.
- Provide a 4-stage progress tracker for order preparation and fulfillment.
- Generate visual dashboards and sales summary reports.
- Identify frequently ordered cake designs and popular menu items.
- Support business owners in making data-driven decisions.
-

Figure 4.1 illustrates the overall workflow of the developed system.

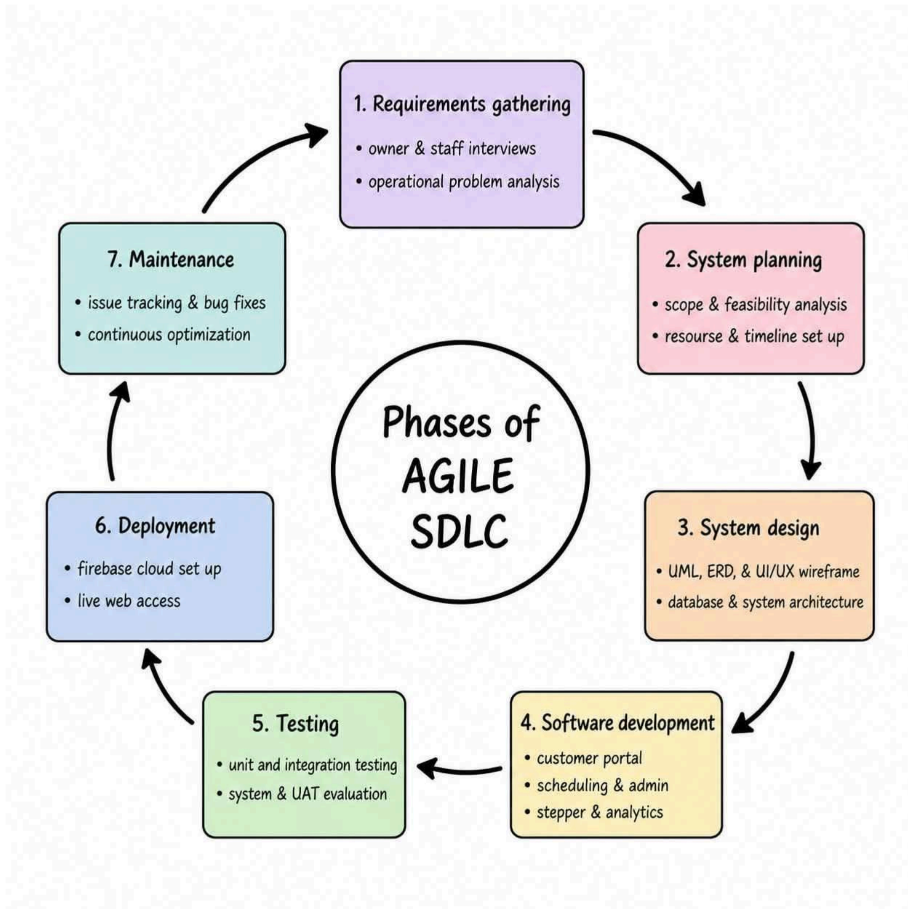
The process begins when a user accesses either the Customer Portal or the Admin Portal. Once logged in, customers can build their custom cakes, check if their desired delivery or pick-up date is available, and place their orders. Authorized staff members then review incoming orders and payment receipts, manage kitchen work using a four-stage progress tracker, adjust schedule limits as needed, and view sales reports generated from completed transactions. All results are presented through simple visual charts and summaries to help the business owner easily understand sales performance.

4.3 System Development Methodology

The development of the Custom Cake Ordering Platform for C&N Bakery followed the Agile Software Development Methodology. Agile was chosen because it allowed the team to build the platform in flexible, step-by-step cycles. This made it easy to make adjustments based on regular feedback from the bakery owner and staff, ensuring the final application met their actual day-to-day needs.

Using Agile, the project was split into short, manageable development cycles called sprints. At the end of each sprint, a working part of the system was built, tested, and reviewed before moving on to the next features. This step-by-step approach helped catch bugs early and reduced overall development risks.

As shown in **Figure 4.2**, the project was carried out across seven key phases:



1. Requirements Gathering

The development process started by collecting information directly from the bakery owner and kitchen staff through interviews and observation. The team looked into common issues with manual paper orders, missed cake specifications, unorganized messaging records, and schedule overbooking. The feedback gathered was used to define the core features and requirements of the platform.

2. System Planning

After identifying what the bakery needed, the team defined the project scope, system features, timeline, and required tools. This phase ensured that the planned application was practical to build within the given timeframe and available technical resources.

3. System Design

In this phase, the overall blueprint of the application was created. The team prepared database structures, system process flows, visual layouts, and module designs. Unified Modeling Language (UML) diagrams, Entity Relationship Diagrams (ERDs), and UI/UX wireframes were created to guide the actual coding stage.

4. Software Development

The coding stage focused on building the application using HTML5, CSS3, and JavaScript for the frontend interface, alongside Google Firebase for real-time database management and authentication. The core features were developed across three main sprints:

- **Sprint 1 (Customer Portal & Builder):** Built the public menu catalog, account registration, the automated custom cake price calculator, and checkout form.
- **Sprint 2 (Smart Scheduling & Admin Controls):** Developed the daily booking calendar limit, the Facebook Messenger payment verification workflow, and the administrative order management dashboard.
- **Sprint 3 (Production Stepper & Sales Analytics):** Built the 4-stage kitchen tracker (Preparing, Baking, Packing, Ready for Pickup/Delivery), automated email status notifications,

customer rating features, and the visual sales reporting dashboard.

5. Testing

Every completed feature was tested individually before being integrated into the full system. Integration testing verified that the frontend screens communicated properly with the Firebase cloud database. Finally, User Acceptance Testing (UAT) was conducted with the bakery owner, kitchen staff, and selected customers to confirm that the platform worked as expected in real-world scenarios.

6. Deployment

After passing all tests, the web application was hosted online to make it accessible to authorized users. Frontend files were published using cloud hosting, while Firebase database security rules were configured to keep administrative controls restricted strictly to C&N Bakery staff. In addition, responsive design testing was performed across mobile phones, tablets, and desktop computers to ensure touch-friendly buttons and proper layout formatting on all screen sizes.

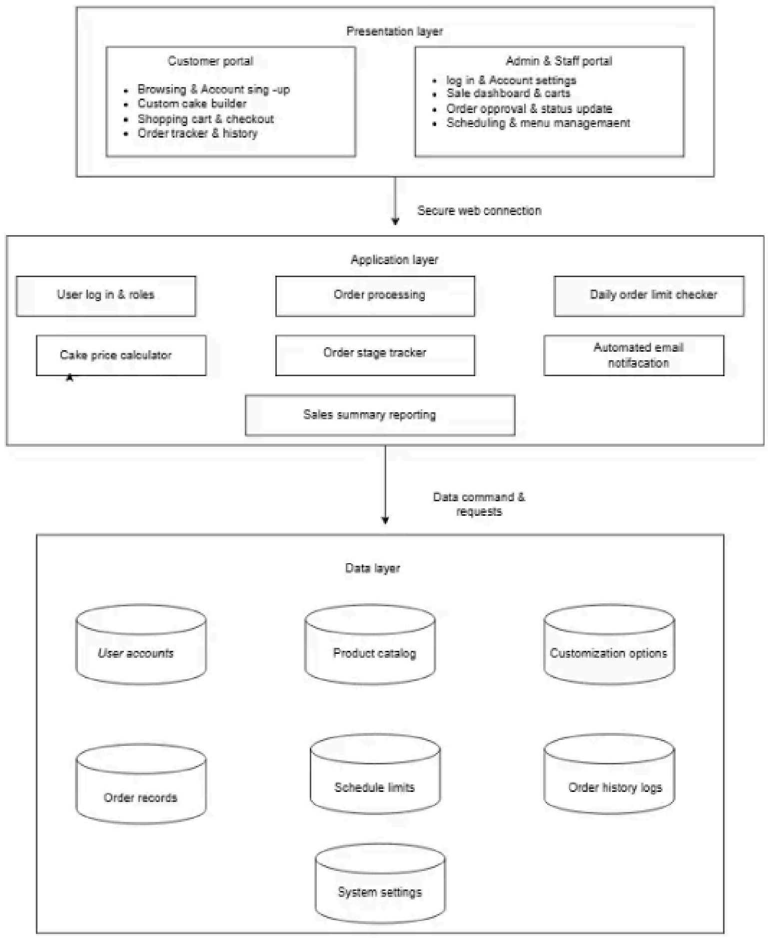
7. Maintenance

During the post-deployment phase, the system was actively monitored to ensure smooth online performance. Small bugs identified during daily use were fixed, database rules were tuned, and minor layout tweaks were made based on ongoing feedback from the bakery staff to maintain system stability.

4.4 System Architecture

The C&N Bakery Web Application uses a Three-Tier Client-Server Architecture. This design divides the system into three simple layers: the presentation layer, the application layer, and the data layer. Organizing the software this way makes the system easier to maintain, scale, and secure.

Figure 4.3 shows the overall architecture of the C&N Bakery ordering and management system.



The system architecture includes the following main parts:

Presentation Layer

The presentation layer is the user interface seen by customers, bakery staff, and administrators. It collects user actions, displays pages, tracks order progress, and shows sales reports.

Users can open the web application on various devices—such as desktop computers, laptops, tablets, and smartphones—using standard web browsers.

The main interface components are:

- Customer Portal
    - Browsing Interface (Home Page, Menu Catalog, FAQ Page, Registration Modal, and Login Modal)
    - Custom Cake Builder and Live Price Estimator
    - Shopping Cart and Order Summary
    - Checkout Page (Pickup or Delivery Date/Time Selection, Daily Order Limit Validation, and Payment Option Selectors for Cash, GCash QR, and Bank Transfer)
    - Account Dashboard (Order Status Tracker, Active Orders List, Order Details Window, Order History with Ratings, and Account Settings)

- Admin and Staff Portal
    - Login Page
    - Dashboard and Sales Overview (Revenue Charts, Sales by Category, Top Products, and Date Filters)
    - Order Management (New Orders Approval Queue and Order Status Controller)
    - Schedule and Capacity Management (Calendar View, Daily Order Limit Editor, and Pickup/Delivery Toggles)

- Menu Management (Product List Editor, Category Manager, Custom Cake Option Settings, and Featured Products Control)
- Order History Archive
- Account and Security Settings

Application Layer

The application layer contains the main logic and rules of the system. It processes user actions, handles logins, checks daily order limits, calculates custom cake prices, sends automated emails, and creates sales summaries for management.

Its main duties include:

- User Authentication and Roles: Manages sign-ups, logins, password changes, and access rights for customers and staff.
- Order and Cart Processing: Checks items in the cart, verifies delivery or pickup choices, applies the 50% downpayment rule for custom cakes, and saves pending orders.
- Schedule and Capacity Validation: Checks requested pickup or delivery dates against daily limits, blocking dates that are already full.

- Custom Cake Pricing Calculation: Calculates real-time total prices based on the customer's selected cake tiers, layer sizes, chiffon flavors, frosting types, and toppers.
- Order Workflow Management: Updates order stages (*Preparing -> Baking -> Packing -> Delivery/Pickup*) and order statuses (*Pending -> Confirmed / Cancelled*).
- Notification Service: Sends automatic email updates to customers whenever staff updates or approves their order status.
- Sales Reporting: Combines order records to show monthly, yearly, and custom-range sales performance.

The application server works as the middle layer, safely processing requests from the website before saving or fetching data from the database.

Data Layer

The data layer stores all system information in a central database. It keeps data organized, reliable, and easy to retrieve.

Stored information includes:

- User Accounts: Customer profiles, admin/staff logins, security tokens, and contact details.

- Product Catalog: Regular cakes, kakanin, pastries, prices, descriptions, photos, and categories.
- Customization Options: Available sizes, flavors, frostings, toppers, and extra costs for custom cakes.
- Order Records: Order details, special customer notes, delivery or pickup methods, selected dates, and payment choices.
- Schedule Settings: Daily order limits and availability toggles for pickup or delivery.
- Order History Logs: Records of order status changes and customer reviews/ratings.
- System Settings: Security rules and website display options.

4.5 Hardware and Software Requirements

The successful operation of the C&N Bakery Custom Cake Ordering Platform relies on specific hardware and software resources. These specifications ensure that the web application runs smoothly, processes custom cake orders without delays, updates calendar limits in real time, and generates sales reports efficiently.

4.5.1 Hardware Requirements

Table 4.1 presents the minimum and recommended hardware specifications required to run and manage the platform. The recommended specs reflect the actual admin environment used by C&N Bakery staff (such as an Acer Aspire laptop with an Intel Core i5 processor and 8 GB of RAM), ensuring quick dashboard loading and responsive data processing.

Table 4.1: Hardware Requirements

Component	Minimum Specification	Recommended Specification (Admin Device)
Processor	Dual-Core 2.0 GHz Processor	12th Gen Intel® Core™ i5-12450H (2.00 GHz) or higher
Memory (RAM)	4 GB RAM	8 GB RAM

Storage	128 GB SSD	512 GB SSD (477 GB usable)
Display / Monitor	1366 \times 768 Resolution	Full HD (1920 \times 1080) Display
Graphics Card	Integrated Graphics	NVIDIA GeForce RTX 2050 (4 GB) / Intel® UHD Graphics
Internet Connection	10 Mbps stable connection	50 Mbps or higher fiber connection
Mobile Device (Client)	Android 10 / iOS 12 or higher	Android 13 / iOS 16 or higher

The recommended hardware setup provides enough processing power to handle daily order traffic, render interactive sales charts, and manage kitchen tracking screens seamlessly.

4.5.2 Software Requirements

Table 4.2 lists the software stack and cloud services used to develop, host, and run the platform. The backend database uses Google Firebase Realtime Database (candn-bakery-default-rtdb.asia-southeast1.firebasedatabase.app), which securely stores and organizes nodes for user profiles, menu categories, custom cake options, daily order limits, and sales records.

Table 4.2: Software Requirements

Software / Tool	**Type / Role**	**Operational Purpose**
Windows 11 Home (64-bit)	Operating System	Serves as the primary operating system environment for system administration and development.

Visual Studio Code	Code Editor	Used as the main software development environment for writing HTML5, CSS3, and Vanilla JavaScript code.
Google Firebase Realtime Database	Cloud Database Management System	Manages cloud data storage across nodes including carts, categories, customization-options, dayFulfillment, dayLimits, featured-cakes, orders, products, reviews, securityQuestion, settings, and users.
HTML5 / CSS3 / Vanilla JavaScript	Web Development Technologies	Used to build the responsive frontend interface, interactive cake pricing engine, and administrative dashboards.

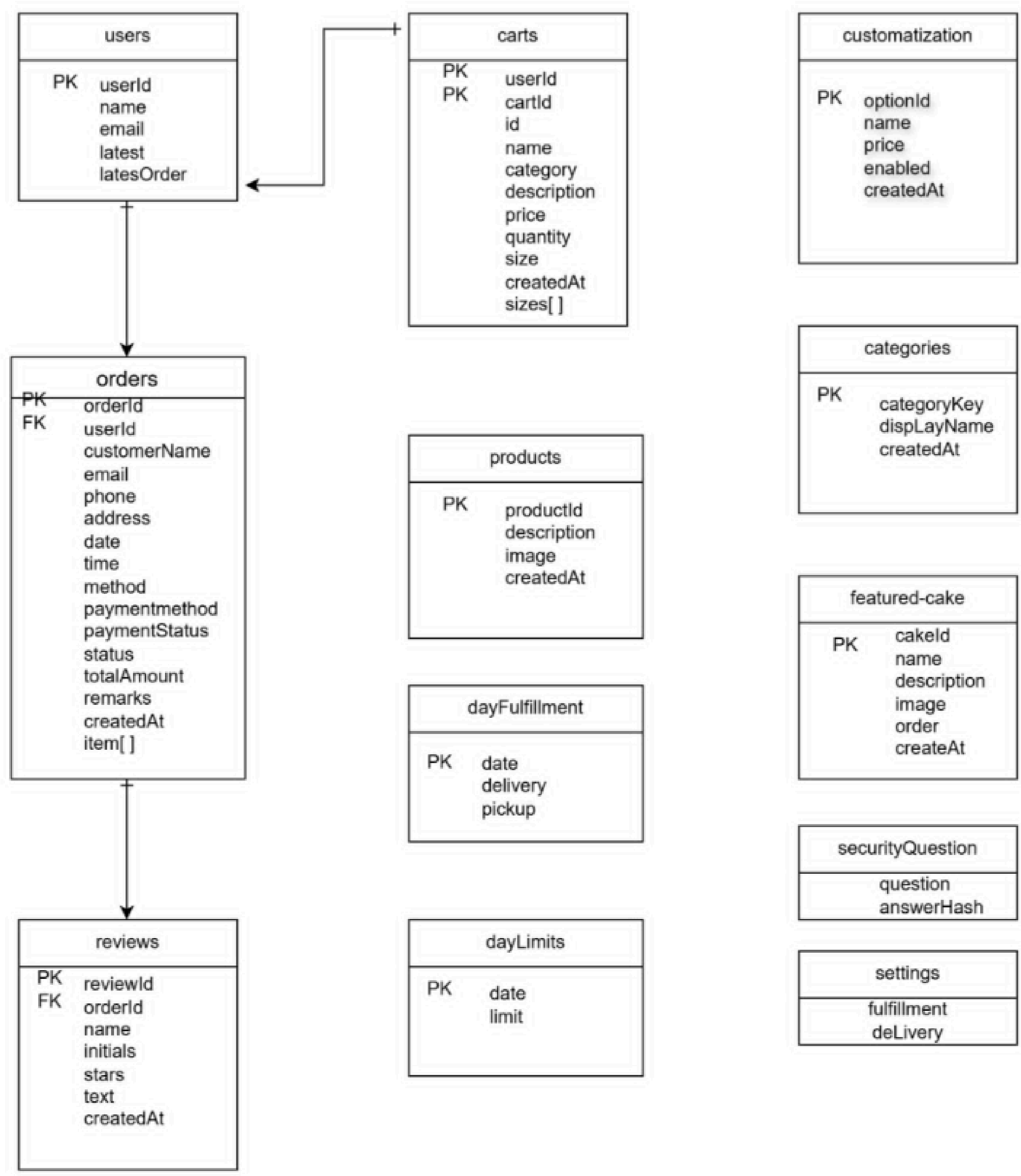
Google Firebase Authenticatio n	Security & Auth Service	Handles secure user sign-ups, customer logins, and administrative permission checks.
Google Chrome / Web Browsers	Client Web Browser	Used by customers and staff to access the web application across desktop and mobile devices.
Git & GitHub	Version Control System	Used to manage source code updates, track code revisions, and maintain system backups during development.

4.6 Database Design

The database serves as the foundation of the C&N Bakery Web Application by organizing and storing customer accounts, menu items, custom cake configurations, active shopping carts, order workflow statuses, store settings, and customer feedback. A well-designed database structure ensures efficient data retrieval, prevents overbooking of custom orders, and maintains overall data integrity.

Rather than using a traditional relational table system (such as MySQL), the platform utilizes **Google Firebase Realtime Database.** In Firebase, data is structured as a hierarchical JSON data tree made up of collection nodes and sub-nodes. This setup allows updates to order statuses, shopping cart changes, and calendar availability to sync instantly between the client interface and the administrative dashboard.

FIGURE 4.4: FIREBASE DATA STRUCTURE DIAGRAM (ERD) OF C&N BAKERY



Primary Database Collection Nodes

The primary database collection nodes include:

- **carts Node**: Temporarily holds active shopping cart items, selected options, prices, and quantities per user.
- **categories Node**: Groups products into distinct menu sections (such as Classic Cakes, Customized Cakes, Kakanin, and Pastries).
- **customization-options Node**: Stores selectable attributes for custom cakes including designs, flavors, frostings, layers, and sizes along with their pricing.
- **dayFulfillment Node**: Controls daily pickup and delivery availability for specific calendar dates.
- **dayLimits Node**: Stores daily maximum order capacities set by management.
- **featured-cakes Node**: Manages showcase cakes rendered on the homepage gallery.
- **orders Node**: Contains detailed transaction logs, fulfillment schedules, payment statuses, ordered items, and 4-stage kitchen preparation states.
- **products Node**: Stores catalog items organized by category.
- **reviews Node**: Holds customer ratings, text feedback, and order reference keys.
- **securityQuestion Node**: Stores administrator password recovery questions and hashed answers.

- **settings Node:** Controls global store configurations such as delivery toggles.
- **users Node:** Stores registered customer accounts, user names, contact emails, and quick order history references.

Detailed Database Node Structures

1. carts Node

This node stores items currently added to a user's active shopping cart prior to checkout.

Field Name	Data Type	Description
userId	Node Key	Unique identifier key matching the user account
cartId	String	Unique shopping cart item key
id	String	Base product item reference key

name	String	Item title (example "Round Caramel Cake (6\")")
category	String	Menu grouping label (e.g., "Classic Cakes")
descript ion	String	Custom item instructions or notes
price	Number	Calculated unit price
quantity	Number	Number of items selected
size	String	Selected cake dimension (example "6\"")
createdA t	Timestam p	Timestamp when the item was added to cart

sizes	Array	Nested array containing available size options and costs

2. categories Node

This node organizes products into main menu sections on the store website.

Field Name	**Data Type**	**Description**
categoryKey	Node Key	Category key (e.g., cakes, customized_cakes, kakanin, pastries)
displayName	String	Category title shown on interface (e.g., "Classic Cakes")
createdAt	Timestamp	Timestamp when the category was created

3. customization-options Node

This node manages custom cake components grouped under sub-nodes for designs, flavors, frostings, layers, and sizes.

Field Name	Data Type	Description
optionId	Node Key	Unique identifier key
name	String	Option label shown to user (e.g., "Printable Topper")
price	Number	Additional cost added to base cake price
enabled	Boolean	Availability toggle (true or false)
createdAt	Timestamp	Timestamp when option was created

4. dayFulfillment Node

This node manages daily fulfillment availability by enabling or disabling delivery and pickup services for specific dates.

Field Name	**Data Type**	**Description**
date	Key (YYYY-MM-DD)	Calendar date (e.g., 2026-07-02)
delivery	Boolean	Service status for home delivery (true or false)
pickup	Boolean	Service status for store pickup (true or false)

5. dayLimits Node

This node limits daily custom order quantities to prevent kitchen overbooking.

Field Name	Data Type	Description
date	Key (YYYY-MM-DD)	Target order date (e.g., 2026-07-04)
limit	Number	Maximum order capacity set by admin (e.g., 3)

6. featured-cakes Node

This node manages showcase cake listings presented on the landing page.

Field Name	Data Type	Description

cakeId	Node Key	Unique featured item key (e.g., -Oszfc36CUTmtLLNoaU_)
name	String	Item name (e.g., "Wedding Cake")
description	String	Written product summary
image	String (Base64)	Encoded image string or link
order	Number	Display priority sequence
createdAt	Timestamp	Post creation timestamp

7. orders Node

This node records submitted customer transactions, payment methods, delivery details, and kitchen stages.

Field Name	**Data Type**	**Description**
orderId	Node Key	Unique transaction key (e.g., -Oy4NTCYrvFiJpTNTLgP)
userId	String	Account key linking to ordering customer
customerName	String	Customer full name
email	String	Customer email address
phone	String	Customer contact number
address	String	Delivery location address or "Store Pickup"

date	YYYY-MM-DD	Chosen pickup or delivery date
time	String	Selected time slot
method	String	Fulfillment type ("pickup" or "delivery")
paymentMethod	String	Payment mode used ("cod", "cop", "gcash")
paymentStatus	String	Current payment state
status	String	Current progress stage ("pending", "preparing", "baking", "packing", "ready")
totalAmount	Number	Total cost of order

remarks	String	Special customer instructions or cancellation notes
createdAt	ISO Timestamp	Order submission timestamp
items	Array	Nested array containing ordered products (stores name, category, price, quantity, description, design, flavor, frosting, layers, message, referencePhoto, and notes)

8. products Node

This node holds standard product catalog items grouped under category sub-nodes (cakes, customized_cakes, kakanin, pastries).

Field Name	**Data Type**	**Description**
productId	Node Key	Unique product record key (e.g., -OwSJvdPhUo4s7x0Bb5f)
description	String	Product specification summary (e.g., "Size: 9\" Design: soft icing")
image	String	Stored product image URL
createdAt	Timestamp	Product addition timestamp

9. reviews Node

This node stores customer feedback and star ratings submitted after order completion.

Field Name	**Data Type**	**Description**

reviewId	Node Key	Unique feedback key (e.g., -OszhMmHQFDQHo-1wqJJ)
orderId	String	Key referencing the completed order
name	String	Reviewer display name
initials	String	First letter initial of reviewer (e.g., "C")
stars	Number	Numerical score awarded (e.g., 4)
text	String	Written review comments
createdAt	ISO Timestamp	Feedback submission timestamp

10. securityQuestion Node

This node stores backend administrative security parameters used for password recovery.

Field Name	Data Type	Description
question	String	Security verification question (e.g., "What city were you born in?")
answerHash	String	Encrypted SHA-256 hash of security answer

11. settings Node

This node controls system-wide operational toggles and global store parameters.

Field Name	Data Type	Description

fulfillment	Object	Sub-node controlling store-wide service availability
delivery	Boolean	Global toggle for enabling/disabling store delivery

12. users Node

This node stores registered customer profiles along with quick-reference links to their latest transactions.

Field Name	**Data Type**	**Description**
userId	Node Key	Unique account ID key (e.g., CztMTS48pwORDEdp6LULReWZPE33)
name	String	Complete customer name

email	String	Registered email address
latest	Object	Quick lookup node for active fulfillment status (orderId, status)
latestOrder	Object	Quick lookup summary node for most recent order (date, orderId, status, total)

CHAPTER 5

CONCLUSION AND RECOMMENDATIONS

5.1 Introduction

This chapter presents the conclusion, summary of findings, and practical recommendations for the Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery. Throughout this research, the project team focused on addressing the operational challenges faced by C&N Bakery, particularly in managing custom order details, preventing kitchen overbooking, tracking multi-stage baking schedules, and analyzing sales performance.

This section synthesizes the system implementation results gathered from Chapter 4, evaluates how well the specific research objectives were met, and outlines actionable recommendations for future developers and business administrators. By combining online ordering, real-time status tracking, automated custom cake price calculations, dynamic daily order limits, and visual sales reporting into a single web platform, the study demonstrates how accessible digital tools can streamline daily operations for local food service enterprises.

5.2 Summary of Findings

The evaluation and implementation of the C&N Bakery Web Application yielded major findings that directly align with the research objectives established at the beginning of the study:

- **Automated Order Processing and Custom Cake Calculations:** The web application successfully replaced manual pencil-and-paper order logs and informal social media messaging threads. The system automatically computes custom cake prices in real time based on user-selected dimensions, tiers, frosting types, and design add-ons. This eliminated miscalculated totals and sped up the order submission process for customers.
- **Effective Prevention of Kitchen Overbooking:** By integrating the dynamic dayLimits and dayFulfillment database modules, the system automatically checks daily order capacities before letting a customer place an order. During testing, selecting dates that reached maximum capacity correctly blocked new orders and prompted users to choose an alternate date. This completely solved the problem of kitchen staff taking on more orders than they could handle on peak days.

- **Clear Order Progress Tracking:** The 4-stage order tracker (Preparing → Baking → Packing → Ready) gave customers clear visibility into their orders without needing to message or call the store repeatedly. Testing confirmed that whenever administrative staff updated an order stage on the management dashboard, the status changed instantly on the customer's order tracking screen.
- **Streamlined Payment and Verification Workflow:** The system provided a smooth flow for both cash transactions and digital payments. Customers who selected digital payments were shown store QR codes and given clear instructions to send proof of payment through Facebook Messenger. Admin review tests showed that staff could quickly confirm receipts, approve orders, and send automated confirmation emails.
- **Actionable Business Intelligence and Feedback:** The visual analytics dashboard successfully aggregated completed order data, converting individual transactions into clear revenue reports and best-selling product charts. Additionally, the integrated rating and review system allowed customers to submit post-order feedback, giving the owner immediate insight into product quality and customer satisfaction.

- **Positive User Evaluation and Acceptance:** System testing with store staff, administrative users, and customers confirmed that the platform is easy to navigate and simple to operate. The clean layout minimized user mistakes and drastically reduced the time needed to train staff on managing incoming orders.

5.3 Conclusion

The Web-Based Order, Tracking, Customization, and Sales Analytics System for C&N Bakery successfully achieved all of its core objectives. It provides a practical and reliable digital solution tailored specifically to the operational needs of a growing local bakery.

Before this system was implemented, C&N Bakery struggled with manual order taking, unorganized custom cake specifications, accidental overbooking during busy seasons, and time-consuming manual sales tallying. The development of this web platform effectively solved these challenges. By moving operations to a centralized Firebase database, the bakery gained a reliable system where custom cake options are priced automatically, daily limits are enforced strictly, and kitchen workflows move forward smoothly.

Furthermore, the integration of real-time status tracking improved customer trust and reduced constant status inquiries. The administrative dashboard simplified daily store management by giving staff full control over catalog management, store settings, daily limit adjustments, and visual sales reports.

In conclusion, this project proves that custom web platforms built with modern tools can make a huge difference for small food businesses. The C&N Bakery Web Application successfully bridges the gap between everyday store operations and simple digital management, improving productivity, reducing manual errors, and helping the business make smarter, data-backed decisions.

5.4 Recommendations

To build on the success of the system and ensure its long-term usefulness for C&N Bakery, the following system improvements, future research directions, and implementation strategies are recommended:

PROPOSED SYSTEM ROADMAP

System Improvements --> In-App Notifications, SMS Alerts, Automated Receipt	
Future Research --> Machine Learning Demand Forecasting, Native App	
Operational Strategy --> Staff Training, Regular Backups, Security Audits	

System Improvements

- **Direct Payment Gateway Integration:** Future developers should consider integrating automated payment gateways (such as PayMaya or Xendit API) directly into the checkout pipeline. This would enable instant payment verification without requiring staff to manually check payment screenshots sent via Facebook Messenger.
- **SMS and In-App Push Notifications:** While the current system uses email notifications, adding direct SMS alerts or browser push notifications would notify customers instantly when their order status changes to Ready for Pickup or Out for Delivery.
- **Automated Low-Stock and Material Alerts:** Expanding the database to track raw ingredients (such as flour, sugar, eggs, and packaging boxes) would allow the system to send

automatic warnings to staff when baking supplies are running low.

- **Multi-Format Report Exporting:** Adding a quick export feature for sales analytics reports—allowing downloads in PDF, CSV, or Excel formats—would make accounting and monthly record-keeping much easier for the bakery owner.

Future Research Directions

- **Predictive Machine Learning for Demand Forecasting**: Future researchers can explore adding machine learning algorithms (such as time-series analysis) to analyze historical sales data. This would help predict high-demand periods and recommend exact inventory levels before major holidays and peak seasons.
- **Longitudinal Impact Study**: Conducting a long-term research study over six to twelve months would help measure the system's ongoing impact on store revenue, operational costs, and overall customer retention rates.
- **Mobile-Native Application Development:** Investigating the development of a companion mobile app for Android and iOS could offer offline order draft features and better push notification support for customers on the go.

Implementation Strategies

- **Comprehensive Staff Training:** C&N Bakery management should run regular training sessions for all kitchen and front-desk staff. Consistent data entry habits—such as updating order stages immediately as work happens in the kitchen—are essential to keep customer tracking accurate.
- **Routine Database Backups and Security Audits:** System administrators should schedule regular database backups and periodically update security rules within Firebase to safeguard customer data and store records.
- **Regular Customer Feedback Reviews:** Business managers are encouraged to check the integrated reviews node regularly to spot quality trends, address customer concerns quickly, and keep product offerings aligned with customer preferences.

5.5 Impact of the Study

This study offers several practical and academic contributions to the field of Information Technology and local business management:

- **Practical Value for Local Enterprises**: It provides a clear, working example of how small bakeries can modernize their ordering, tracking, and customization processes without relying on expensive, overly complex software.

- **Practical NoSQL Database Blueprint:** It offers a real-world case study on using document-oriented NoSQL structures (Firebase Realtime Database) to handle customized product options, denormalized order snapshots, dynamic calendar limits, and real-time status updates.
- **Better Customer Experience:** It improves customer satisfaction by making custom cake selection transparent, providing real-time order visibility, and removing the frustration of last-minute order cancellations due to kitchen overbooking.
- **Data-Driven Operations:** It shifts the bakery away from guesswork by converting everyday transaction logs into clear visual charts, giving the business owner the data needed to plan inventory, highlight popular products, and run targeted promotions.

5.6 Chapter Summary

Chapter 5 concluded the study by summarizing the core findings, evaluating project achievements, and providing practical recommendations for the C&N Bakery Web Application. The findings showed that automating order entry, enforcing daily order limits, offering step-by-step order tracking, and providing clear sales analytics significantly improved store operations.

By addressing the real-world operational challenges of C&N Bakery, this research offers a solid foundation for future developers to build upon while helping small food enterprises embrace digital tools for long-term growth.